Voca-Twin

FINAL YEAR PROJECT

Submitted by

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FINAL YEAR PROJECT PHASE-1 DOCUMENTATION

STATEMENT OF SUBMISSION

Submitted to the University of Lahore in partial fulfillment of the requirement for the award of degree of Bachelors of Science in Software Engineering (BSSE)

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ABSTRACT

Voca-Twin is an AI-powered virtual assistant system that revolutionizes human-computer interaction by integrating real-time voice cloning, 3D avatar customization, and multilingual support. Leveraging advanced technologies like TensorFlow and Unity, it delivers personalized and interactive experiences through emotion-aware responses, seamless real-time communication, and adaptive interactions. Users can clone their voices, customize avatars for visual representation, and communicate in multiple languages, including regional dialects like Urdu. The system also supports accessibility for speech-impaired users through text-based inputs and enables participation in virtual meetings with AI-enhanced voice capabilities. Designed for applications such as customer support and personalized chatbots, Voca-Twin sets a new standard for inclusive, empathetic, and engaging virtual assistant solutions.

DEDICATION

First and foremost, we dedicate this document to ALLAH Almighty, who has bestowed upon us the ability to work diligently. He is the source of our wisdom, knowledge, and understanding. Second, we dedicate it to our parents, who have never failed to pray for us and send us good wishes. Whatever we achieve in life is due to their unwavering support, prayers, and advice. Lastly, we dedicate this work to our teacher, **Mr. Najaf Ali**, who has been a great mentor and guide throughout the project. His motivation and assistance have been invaluable to us.

ACKNOWLEDGMENT

Our heartfelt gratitude to Allah Almighty, who has provided us with the wisdom, strength, and perseverance to complete this project, even in the most challenging circumstances. We extend our deepest thanks to our parents, whose unwavering support, prayers and sacrifices have laid the foundation for our education and success.

We would also like to express our sincere gratitude to all the professors who have guided and supported us throughout this journey. Our profound thanks go to our supervisor, **Mr. Najaf Ali**, for his invaluable mentorship, encouragement and assistance in every step of this project. Special thanks to our friends and seniors for their advice and support during this endeavor.

This project, **Voca-Twin** represents our effort to create an innovative AI-powered virtual assistant that integrates real-time voice cloning, 3D avatar customization and multilingual support, aiming to redefine human-computer interaction and make it more inclusive, empathetic and engaging.

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CHAPTER 1: INTRODUCTION TO THE PROBLEM

1.1 Introduction

In the fast-paced world of artificial intelligence (AI) and virtual assistance, existing solutions like Siri, Alexa, and Google Assistant have revolutionized human-computer interaction. However, they remain limited in their ability to provide **deep personalization**, **emotional intelligence**, and **cultural adaptability**. These limitations result in impersonal, robotic interactions that fail to resonate with users on a personal level.

With increasing reliance on AI-driven tools for communication, productivity, and accessibility, there is a critical need for a solution that can replicate human-like interactions. **Voca-Twin: The Virtual Voice Companion** addresses this gap by integrating advanced **voice cloning**, **avatar customization**, and **emotionally intelligent AI** into a unified mobile application. Voca-Twin empowers users to create their digital twins—virtual companions that replicate their voice, visual appearance, and emotional tone, enabling hyper-personalized, real-time interactions.

The project brings together cutting-edge technologies like **natural language processing (NLP)**, **AI-driven voice synthesis**, and **3D/real-life avatars** to deliver a transformative user experience. From enhancing accessibility for differently-abled individuals to revolutionizing business communication and education, Voca-Twin has the potential to redefine the virtual assistant landscape.

1.2 Purpose

The purpose of **Voca-Twin** is to bridge the gap between human-like AI interactions and the impersonal nature of current virtual assistants. This project aims to:

1. Provide a **personalized AI companion** capable of replicating a user's voice and appearance for realistic, relatable interactions.
2. Introduce **emotionally intelligent AI responses** that adapt to user tone, intent, and emotional context.
3. Support **regional languages** like Urdu to address cultural gaps and enhance inclusivity for underrepresented demographics.
4. Improve **accessibility** for individuals with disabilities or limited technical expertise by offering intuitive, voice-based interactions.
5. Offer businesses a scalable, AI-powered tool for **customer engagement**, **education**, and **productivity**, fostering deeper trust and connection.

1.3 Objectives

The key objectives of **Voca-Twin** are:

1. **Voice Cloning**: Develop an AI-powered module that accurately replicates a user's voice, capturing tone, pitch, and emotional nuances.
2. **Avatar Customization**: Implement a system that allows users to create 3D or real-life avatars resembling their appearance, offering full customization options.
3. **Emotionally Intelligent AI**: Integrate advanced **natural language processing** (NLP) and AI models to generate intelligent, emotion-aware responses.
4. **Multilingual Support**: Enable seamless communication in multiple languages, including Urdu, to cater to diverse user bases.
5. **Real-Time Interaction**: Deliver low-latency, real-time responses for smooth, dynamic conversations.

1.4 Existing Solution

The current virtual assistant market is dominated by solutions like **Google Assistant**, **Amazon Alexa**, **Apple Siri**, and niche tools like **Descript's Overdub** and **Synthesia**. While these solutions have made significant strides in AI-driven communication, they exhibit critical shortcomings:

Key Limitations of Existing Solutions:

1. **Lack of Personalization**: Current systems offer generic, pre-programmed responses that fail to adapt to individual user preferences.
2. **Absence of Visual Representation**: Solutions like Google Assistant and Alexa lack avatars, limiting emotional engagement.
3. **No Emotional Intelligence**: Responses are robotic, lacking the ability to adapt to emotional tone or intent.
4. **Cultural Barriers**: Regional languages like Urdu are unsupported, excluding large user demographics.
5. **Limited Real-Time Interaction**: Voice cloning tools like Overdub focus on creative content, not dynamic, real-time conversations.

1.5 Proposed Solution

Voca-Twin is a revolutionary mobile application that overcomes the limitations of existing virtual assistants through an integrated, feature-rich solution. The project combines **voice cloning**, **3D/real-life avatars**, and **emotionally intelligent AI** to deliver hyper-personalized, human-like interactions in real time.

Key Features of Voca-Twin:

1. **AI Voice Cloning**: Unlike existing tools, Voca-Twin replicates the user's voice in real time, capturing emotional nuances and speaking style for highly realistic interactions.
2. **3D/Real-Life Avatar Customization**: Users can create dynamic avatars that reflect their appearance, enhancing visual engagement. Unlike **Synthesia**, these avatars are interactive and not limited to static videos.
3. **Emotionally Intelligent AI**: Leveraging advanced **NLP** and **AI models**, Voca-Twin generates responses that adapt to the user's tone, intent, and emotional state—a feature missing in Google Assistant and Alexa.
4. **Multilingual Support**: By supporting languages like **Urdu**, Voca-Twin bridges cultural gaps and provides inclusive AI experiences for underrepresented regions.
5. **Real-Time Interaction**: Voca-Twin ensures low-latency, dynamic communication, unlike Overdub and Synthesia, which focus on pre-recorded or static outputs.

Why Voca-Twin is Better:

- **Personalization**: Combines voice cloning and avatar customization to deliver a truly human-like AI experience.
- **Emotional Intelligence**: Generates adaptive, emotion-aware responses, unlike competitors with robotic interactions.
- **Inclusivity**: Supports regional languages like Urdu, making it accessible to diverse user bases.
- **Real-Time Engagement**: Provides seamless, low-latency communication for dynamic, real-time interactions.

By addressing these gaps, **Voca-Twin** emerges as a superior, future-ready solution that redefines AI-powered virtual assistants. It is not just a tool but a **companion** that understands, responds, and engages with users on a deeply personal level.

1.6 Gap Analysis

The following table summarizes the gaps in existing solutions and how Voca-Twin addresses them:

Table 1: Gap Analysis

Key Gap	**Existing Solutions**	**Voca-Twin Solution**
Lack of Personalization	Generic, pre-programmed responses	Real-time AI voice cloning and avatar customization
Absence of Visual Representation	No avatars or static video avatars	Interactive 3D/real-life avatars
No Emotional Intelligence	Robotic responses without adaptation	Emotion-aware, adaptive AI responses
Cultural and Language Barriers	Limited support for regional languages	Multilingual support, including Urdu
Limited Real-Time Interaction	Static or delayed interactions	Low-latency, dynamic real-time communication
Accessibility for All Users	Complex interfaces; limited usability	Voice-based, intuitive controls for all skill levels

With its innovative features and holistic approach, **Voca-Twin** stands out as a groundbreaking solution that transforms virtual assistants into intelligent, empathetic, and inclusive companions, ready to meet the demands of modern users.

CHAPTER 2: SOFTWARE REQUIREMENT SPECIFICATION (SRS)

2.1 Introduction

2.1.1 Purpose

The purpose of this **Software Requirement Specification (SRS)** is to:

1. Define the functional and non-functional requirements of the **Voca-Twin** system.
2. Serve as a reference for developers, testers, and stakeholders to ensure project alignment.
3. Provide a foundation for system design, implementation, and validation.

Intended Audience: This document is intended for the project team, evaluators, developers, testers, and other stakeholders involved in the project lifecycle.

2.1.2 Scope

The **Voca-Twin** system will:

1. Replicate user voices through **AI-driven voice cloning**.
2. Provide **3D avatar customization** for visual representation.
3. Support **real-time interaction** with low latency.
4. Enable **multilingual communication**, including regional languages like Urdu.
5. Deliver a seamless experience across multiple platforms (Android, iOS, Web).
6. Ensure data security through encryption and user authentication mechanisms.

The system will **not**:

- Support physical hardware integration beyond standard devices (e.g., smartphones, tablets, and PCs).
- Replace existing enterprise-level AI tools entirely but will complement them for personalized experiences.

2.1.3 Definitions, Acronyms, and Abbreviations

- **AI**: Artificial Intelligence
- **TTS**: Text-to-Speech
- **NLP**: Natural Language Processing
- **FR**: Functional Requirement
- **NFR**: Non-Functional Requirement
- **UI**: User Interface
- **UX**: User Experience
- **3D**: Three-Dimensional

2.1.4 References

1. Google Assistant Documentation: https://assistant.google.com
2. Amazon Alexa Developer Guide: https://developer.amazon.com/alexa
3. Unity Documentation for 3D Rendering: https://docs.unity3d.com
4. TensorFlow for Voice Cloning: https://www.tensorflow.org

2.1.5 Overview

This Software Requirements Specification (SRS) document provides a detailed description of the functional and non-functional requirements for the software product. It outlines the system's objectives, features, and design considerations, offering a comprehensive guide for the development and evaluation of the product. The document includes the following sections:

- **Introduction**: Establishes the purpose, scope, and definitions relevant to the software product.
- **Overall Description**: Provides a high-level view of the product, including its context, system interfaces, user interfaces, hardware/software dependencies, and communication protocols.
- **Product Functions**: Details the functional requirements of the system, describing each module and its expected behavior.
- **User Characteristics**: Defines the target user base and their general characteristics.
- **Constraints**: Lists any limitations that may impact development, including regulatory policies, hardware limitations, and other operational factors.
- **Assumptions and Dependencies**: Identifies external factors that may influence the requirements.
- **Apportioning of Requirements**: Highlights any requirements that may be deferred to future versions of the system.

2.2 Overall Description

2.2.1 Product Perspective

Voca-Twin is a standalone mobile application that integrates **AI voice cloning** and **3D or real-life avatars** with advanced NLP capabilities for intelligent, emotional responses. The product is independent and self-contained, designed for deployment across Android, iOS, and web platforms.

System Interfaces:

- TensorFlow for AI-driven voice cloning.
- Unity or Unreal Engine for 3D avatar rendering.
- Firebase for real-time database management and authentication.
- OpenAI GPT API for intelligent, emotion-aware chatbot responses.

User Interfaces:

- Intuitive screens for voice recording, avatar customization, and real-time chat interactions.

Hardware Interfaces:

- Compatible with devices running Android, iOS or modern web browsers.

Software Interfaces:

- TensorFlow, Unity/Unreal Engine, Firebase and OpenAI GPT integration.

Communications Interfaces:

- HTTPS for secure data transmission.
- WebSockets for real-time interactions.

Memory:

- Minimum: 4GB RAM and 500MB storage for application installation.

Operations:

- Backup and recovery mechanisms for user data.

Site Adaptation Requirements:

- Cloud-based deployment ensures scalability and minimal on-site adaptation.

2.2.2 Product Functions

Below in functional requirement 2.3.1.

2.3 Specific Requirements

2.3.1 Functional Requirements

The following table outlines the functional requirements (FRs) of the Voca-Twin system:

Table 2: Functional Requirements

ID	Name	Description	Input	Output	Basic Workflow
FR01	User Authentication	Allows users to securely create an account (Sign-Up) or log in to the system (Sign-In).	Email and password	Access to system features	1. User enters credentials. 2. System validates. 3. Access granted or denied.
FR02	Voice Cloning	Allows users to replicate their voice for personalized interactions.	User audio samples	Cloned user voice	1. User records voice. 2. System processes input. 3. Voice model is generated.
FR03	Real-Time Interaction	Processes user input and generates responses instantly.	Voice or text input	Real-time response	1. User inputs data 2. System processes. 3. Response is delivered instantly.
FR04	Text-to-Speech Conversion	Converts user- provided text into speech using the cloned voice.	Text input	Speech output	1. User inputs text 2. System processes. 3. Speech is played back.
FR05	AI-Powered Personalized Chatbot	Provides conversational responses to user queries using the cloned voice.	Conversational query	Response in cloned voice	1. User inputs query. 2. System processes. 3. Response delivered in cloned voice.
FR06	Accessibility for Speech-Impaired Users	Enables speech-impaired users to provide input through text and receive responses in their cloned voice.	Text input	Cloned voice response	1. User provides input. 2. System processes. 3. Response is delivered in cloned voice.

FR07	Multilingual Support	Supports multiple languages, including regional dialects like Urdu.	Voice/text input (language)	Language-specific response	1. System detects input language. 2. Generates localized output.
FR08	Virtual Meeting Participation	Enables users to join virtual meetings and use their cloned voice for interaction.	Voice input	Cloned voice heard by participants	1. User speaks. 2. System generates cloned voice. 3. Participants hear the cloned voice.
FR09	Automated Voice Notes and Reminders	Allows users to set voice-based reminders using their cloned voice.	Reminder details	Reminder played in cloned voice	1. User sets reminder. 2. System schedules. 3. Reminder plays at specified time.
FR10	Avatar Customization	Enables users to create and customize their digital avatars.	User customization preferences	Custom 3D avatar	1. User selects preferences. 2. System renders avatar. 3. Avatar is saved.
FR11	Interactive Voice-Based Customer Support	Provides AI-powered customer support using the cloned voice for interaction.	Customer queries	Cloned voice responses	1. User inputs query. 2. System retrieves response. 3. Response delivered in cloned voice.
FR12	Cross-Platform Support	Ensures consistent user experience across Android, iOS, and web platforms.	User login/device access	Unified user experience	1. User logs in. 2. System synchronizes data. 3. Interface adjusts for platform.
FR13	User Feedback Mechanism	Allows users to provide feedback for system improvement.	User feedback data	Feedback stored in database	1. User submits feedback. 2. System stores data. 3. Feedback used for improvements.

2.3.2 Non-Functional Requirements

The non-functional requirements (NFRs) ensure the system's quality, performance, and reliability:

Table 3: Non-Functional Requirement

Category	Requirement	Description
Usability	Intuitive User Interface	The system must provide an easy-to-use, accessible UI for all users.
Reliability	High Uptime	Ensure 99.9% system availability with redundancy and failover mechanisms.
Performance	Low Latency	System must respond within 2.5 seconds for real-time interactions.
Security	Data Encryption	Voice data and user profiles must be encrypted to prevent unauthorized access.
Portability	Cross-Platform Compatibility	The system must function seamlessly on Android, iOS, and web platforms.
Maintainability	Scalable Architecture	The system must support scalability for growing user bases and data volumes.
Design	Modular Design	Follow a modular design to allow easy updates and feature enhancements.
Compliance	Data Protection Regulations	Ensure compliance with GDPR and similar data protection laws.

CHAPTER 3: USE CASE ANALYSIS

3.1 Individual Use Case Diagrams

This section provides detailed use case diagrams that illustrate the core functionalities of the system. Each diagram highlights how users interact with specific features of the project, showcasing the flow of actions and the system's responses. These diagrams are essential for understanding the operational structure and user interaction within the system.

3.1.1 Use Case Diagram for UC_01: User Authentication (Sign-Up and Sign-In)

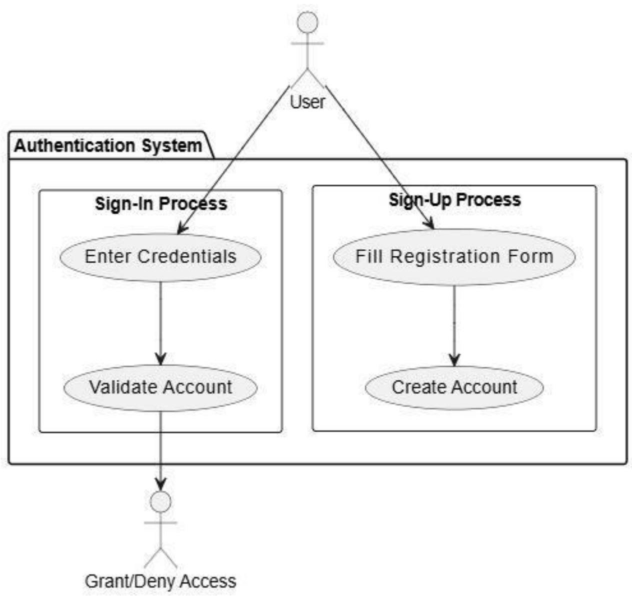


Figure 1: Use Case Diagram for UC_01: User Authentication (Sign-Up and Sign-In)

Table 4: Use Case Table for UC_01: User Authentication (Sign-Up and Sign-In)

Use Case ID	**UC_01**
Use Case Name	User Authentication (Sign-Up and Sign-In)
Description	Allows users to securely create an account (Sign-Up) or log in to the system (Sign-In). Authentication ensures authorized access to system functionalities.
Primary Actor	User
Secondary Actor	System
Pre-Condition	1. User provides valid credentials (email and password) or creates a new account.
Post- Condition	User is authenticated and gains access to system features.
Basic Flow	1. User enters email and password (Sign-In) or fills out the registration form (Sign-Up). 2. System validates the credentials or creates a new account. 3. User is granted access to the system.
Alternate Flow	If credentials are invalid, the system prompts the user to re-enter or recover the password.

3.1.2 Use Case Diagram for UC_02: Real-Time Voice Cloning

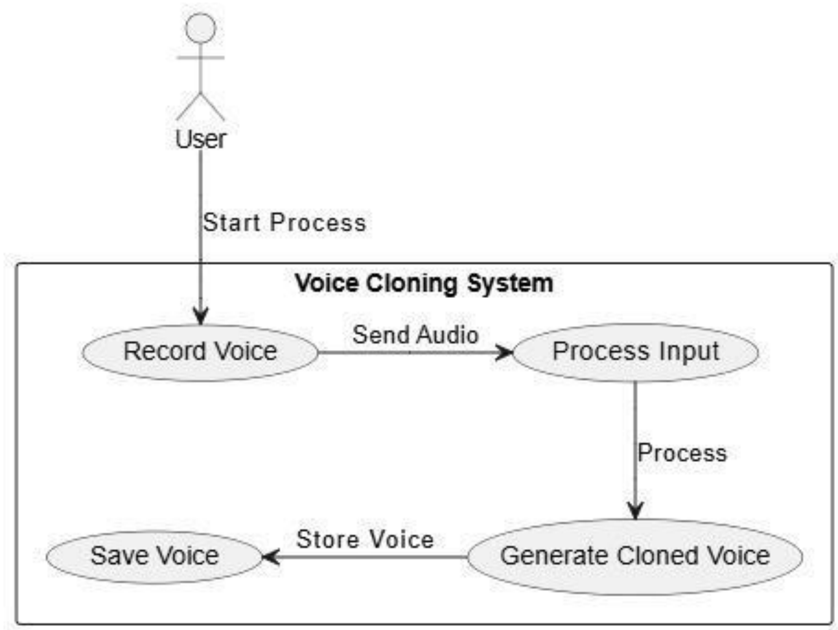


Figure 2: Use Case Diagram for UC_02: Real-Time Voice Cloning

Table 5: Use Case Table for UC_02: Real-Time Voice Cloning

Use Case ID	UC_02
Use Case Name	Real-Time Voice Cloning
Description	Enables users to clone their voice in real time for seamless interaction. The system processes the user's voice input to generate a cloned voice that matches tone, accent, and timing.
Primary Actor	User
Secondary Actor	System
Pre- Condition	1. User's voice data is available. 2. Microphone and audio processing hardware are functional.
Post-Condition	A cloned voice is generated and saved for real-time use.
Basic Flow	1. User records a voice input. 2. System processes the input and generates the cloned voice. 3. The cloned voice is played back or saved for future interactions.
Alternate Flow	If the input is invalid or noisy, the system prompts the user to re-record.

3.1.3 Use Case Diagram for UC_03: Real-Time Interaction

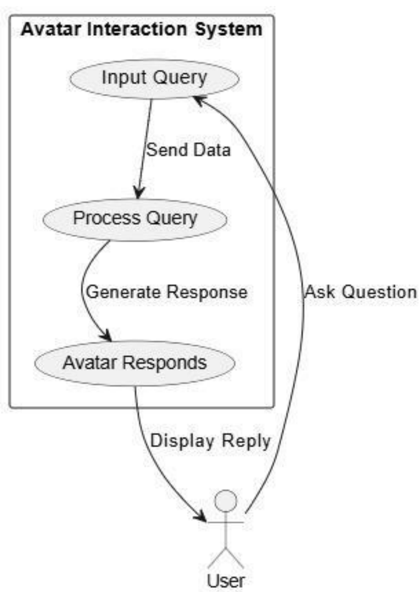


Figure 3: Use Case Diagram for UC_03: Real-Time Interaction

Table 6: Use Case Table for UC_03: Real-Time Interaction

Use Case ID	**UC_03**
Use Case Name	Real-Time Interaction
Description	Processes user input and generates responses instantly, ensuring seamless interaction.
Primary Actor	User
Secondary Actor	System
Pre-Condition	1. System is active and ready to process input. 2. User is authenticated.
Post-Condition	User receives a real-time response to their input.
Basic Flow	1. User provides input (text or voice). 2. System processes the input. 3. System delivers a real-time response.
Alternate Flow	If the system fails to process the input, it prompts the user to retry.

3.1.4 Use Case Diagram for UC_04: Text-to-Speech Conversion

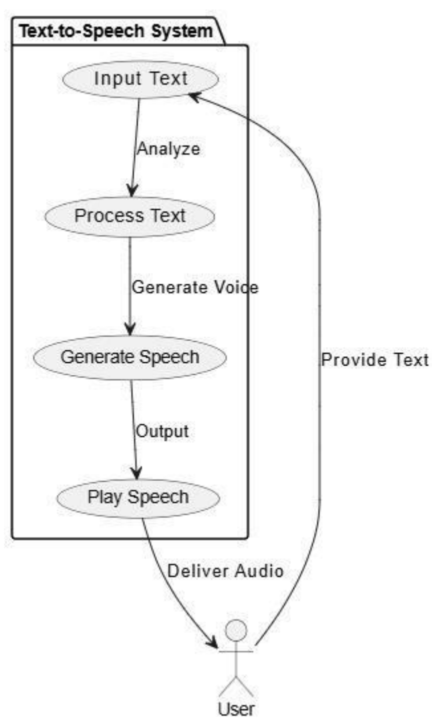


Figure 4: Use Case Diagram for UC_04: Text-to-Speech Conversion

Table 7: Use Case Table for UC_04: Text-to-Speech Conversion

Use Case ID	**UC_04**
Use Case Name	Text-to-Speech Conversion
Description	Converts user-provided text into speech using the cloned voice, ensuring natural tone and clarity.
Primary Actor	User
Secondary Actor	System
Pre-Condition	1. Text-to-speech engine is functional. 2. Cloned voice is pre-saved.
Post- Condition	Input text is converted to speech and played back accurately.
Basic Flow	1. User inputs text. 2. System processes the text and generates speech using the cloned voice. 3. Speech is played back to the user.
Alternate Flow	If the input text is too long, the system splits it into manageable parts and processes them sequentially.

3.1.5 Use Case Diagram for UC_05: Text-to-Speech Conversion

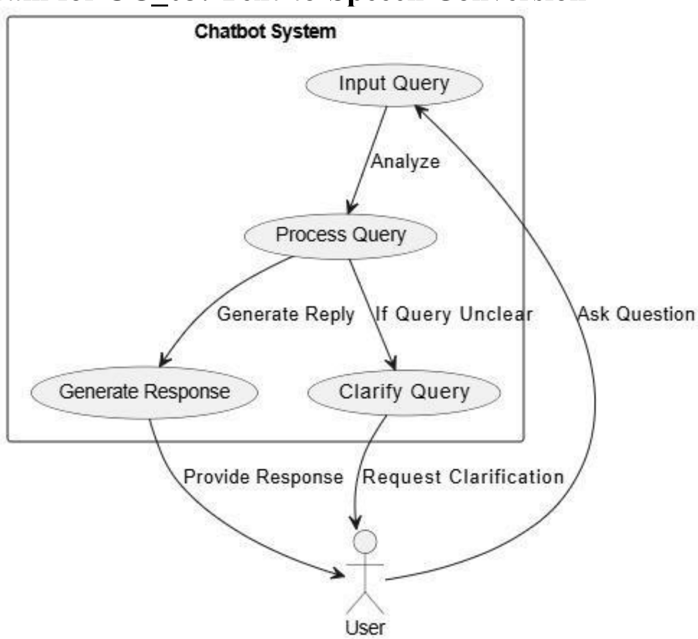


Figure 5: Use Case Diagram for UC_05: AI-Powered Personalized Chatbot

Table 8: Use Case Table for UC_05: AI-Powered Personalized Chatbot

Use Case ID	**UC_05**
Use Case Name	AI-Powered Personalized Chatbot
Description	Provides conversational responses to user queries using the cloned voice.
Primary Actor	User
Secondary Actor	System
Pre-Condition	1. Chatbot AI is integrated with the cloned voice engine. 2. Query data is available.
Post-Condition	Responses are accurate, timely, and spoken in the cloned voice.
Basic Flow	1. User inputs a conversational query. 2. System processes the query and generates a response. 3. Response is delivered in the cloned voice.
Alternate Flow	If the query is unclear, the system prompts the user for clarification.

3.1.6 Use Case Diagram for UC_06: Accessibility for Speech-Impaired Users

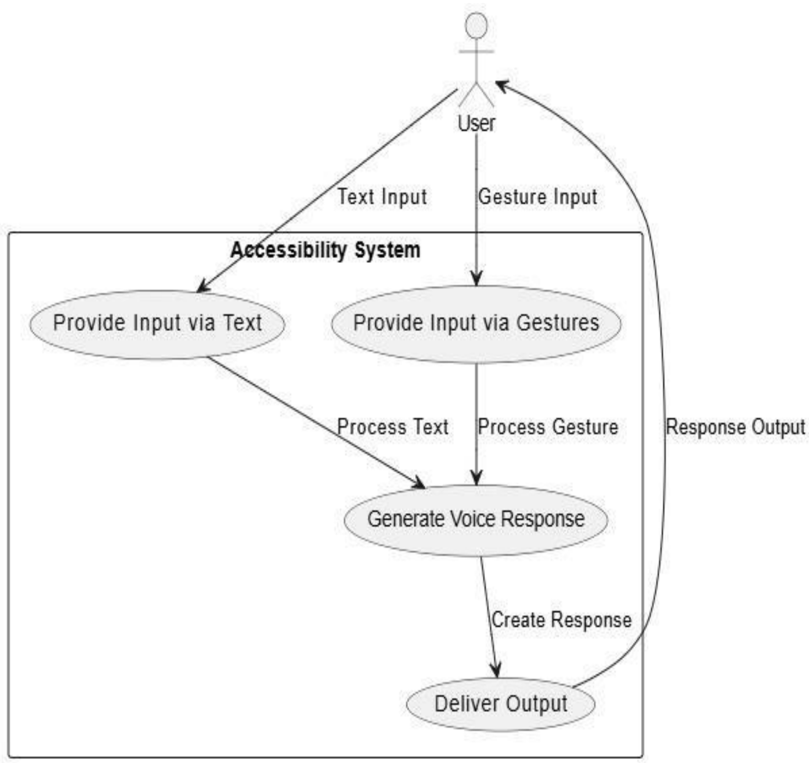


Figure 6: Use Case Diagram for UC_06: Accessibility for Speech-Impaired Users

Table 9: Use Case Table for UC_06: Accessibility for Speech-Impaired Users

Use Case ID	**UC_06**
Use Case Name	Accessibility for Speech-Impaired Users
Description	Allows speech-impaired users to provide input through alternative methods (e.g., text) and receive responses in their cloned voice.
Primary Actor	User
Secondary Actor	System
Pre-Condition	1. User's voice data is pre-recorded. 2. Alternative input methods are active.
Post- Condition	System produces accurate responses in the user's cloned voice.
Basic Flow	1. User provides input through text. 2. System processes the input and generates a voice response. 3. Response is delivered in the user's cloned voice.
Alternate Flow	If input is unclear, the system prompts the user for clarification.

3.1.7 Use Case Diagram for UC_07: Multilingual Support

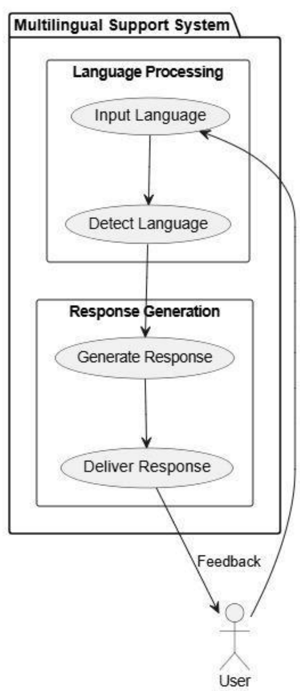


Figure 7: Use Case Diagram for UC_07: Multilingual Support

Table 10: Use Case Table for UC_07: Multilingual Support

Use Case ID	UC_07
Use Case Name	Multilingual Support
Description	Enables users to interact with the system in multiple languages, with responses provided in the cloned voice.
Primary Actor	User
Secondary Actor	System
Pre-Condition	1. Multilingual text-to-speech engine is active. 2. Supported languages are configured.
Post-Condition	System responds accurately in the user's cloned voice and the correct language.
Basic Flow	1. User inputs text or voice in a specific language. 2. System detects the language and generates a response. 3. Response is delivered in the cloned voice.
Alternate Flow	If the input language is unsupported, the system defaults to a primary language and informs the user.

3.1.8 Use Case Diagram for UC_08: Virtual Meeting Participation

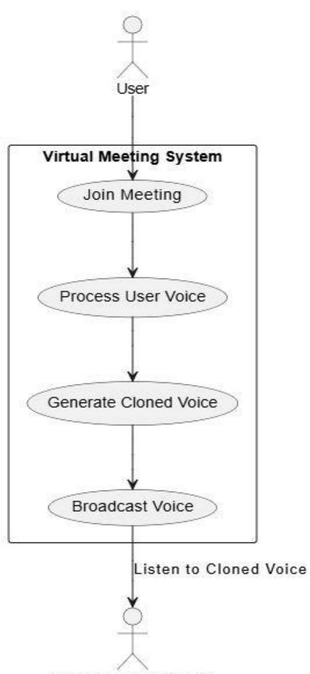


Figure 8: Use Case Diagram for UC_08: Virtual Meeting Participation

Table 11: Use Case Table for UC_08: Virtual Meeting Participation

Use Case ID	**UC_08**
Use Case Name	Virtual Meeting Participation
Description	Enables users to join virtual meetings and use their cloned voice for interaction.
Primary Actor	User
Secondary Actor	System
Pre-Condition	1. Stable internet connection. 2. User's microphone and speaker are active.
Post-Condition	Cloned voice is audible and clear to other participants.
Basic Flow	1. User joins a virtual meeting. 2. User speaks, and the system generates the cloned voice output. 3. Other participants hear the cloned voice clearly.
Alternate Flow	If the internet connection is unstable, the system notifies the user and retries.

3.1.9 Use Case Diagram for UC_09: Automated Voice Notes and Reminders

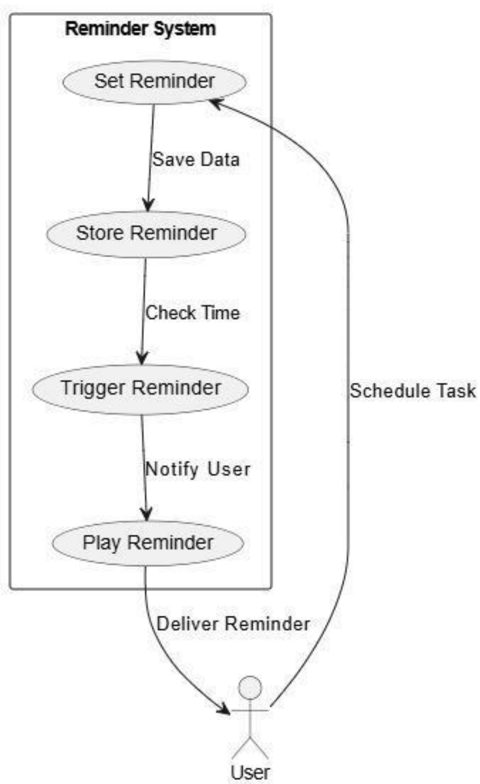


Figure 9: Use Case Diagram for UC_09: Automated Voice Notes and Reminders

Table 12: Use Case Table for UC_09: Automated Voice Notes and Reminders

Use Case ID	**UC_09**
Use Case Name	Automated Voice Notes and Reminders
Description	Allows users to set voice-based reminders using their cloned voice.
Primary Actor	User
Secondary Actor	System
Pre-Condition	1. User's voice clone is saved. 2. Reminder application is integrated with Voca-Twin.
Post-Condition	Reminder plays in the cloned voice at the specified time.
Basic Flow	1. User sets a voice reminder. 2. System schedules the reminder. 3. Reminder is played in the cloned voice at the set time.
Alternate Flow	If the reminder cannot be set, the system notifies the user and retries.

3.1.10 Use Case Diagram for UC_10: Avatar Creation and Personalization

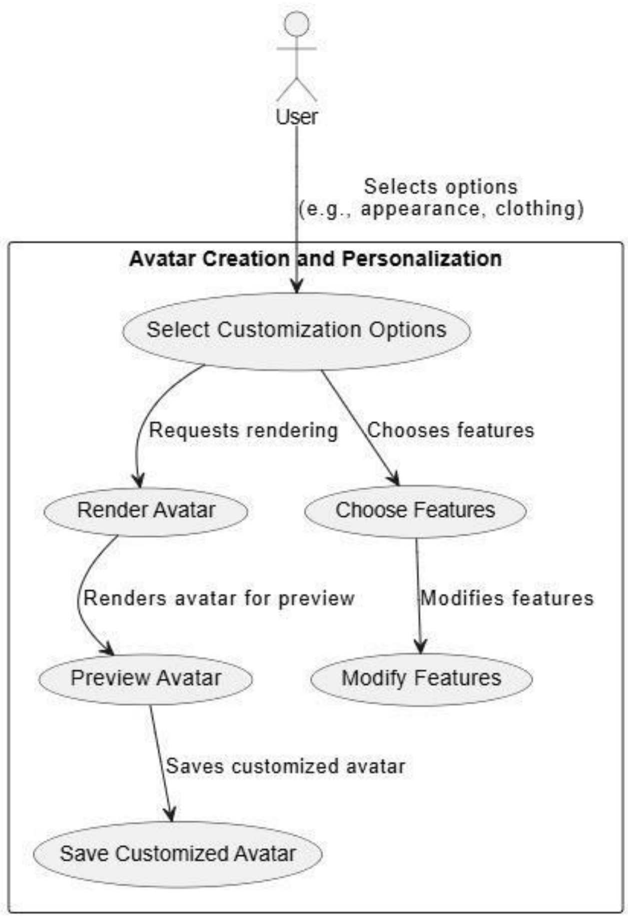


Figure 10: Use Case Diagram for UC_10: Avatar Creation and Personalization

Table 13: Use Case Table for UC_10: Avatar Creation and Personalization

Use Case ID	UC_10
Use Case Name	Avatar Creation and Personalization
Description	Enables users to create and customize their digital avatars with personalized features.
Primary Actor	User
Secondary Actor	System
Pre-Condition	Avatar customization module is active, and the user is authenticated.
Post-Condition	Customized avatar is saved for future use.
Basic Flow	1. User selects customization options (e.g., appearance, clothing). 2. System renders the avatar based on the user's preferences. 3. Avatar is saved in the database.
Alternate Flow	If customization fails, the system notifies the user and prompts for retry.

3.1.11 Use Case Diagram for UC_11: Interactive Voice-Based Customer Support

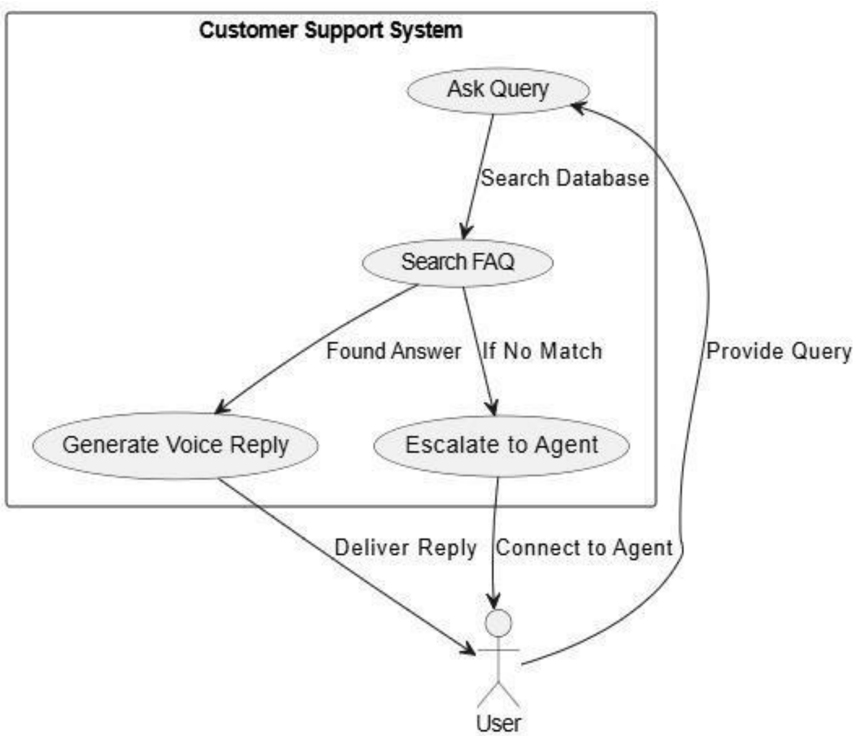


Figure 11:Use Case Diagram for UC_11: Interactive Voice-Based Customer Support

Table 14: Use Case Table for UC_11: Interactive Voice-Based Customer Support

Use Case ID	**UC_11**
Use Case Name	Interactive Voice-Based Customer Support
Description	Provides AI-powered customer support using the cloned voice for interaction.
Primary Actor	User
Secondary Actor	System
Pre-Condition	1. FAQ database is integrated. 2. Voice cloning and response system are active.
Post-Condition	Cloned voice responds accurately and conversationally to customer queries.
Basic Flow	1. User inputs a query. 2. System retrieves the relevant FAQ and generates a response. 3. Response is delivered in the cloned voice.
Alternate Flow	If the query is not found, the system escalates it to a human agent.

3.1.12 Use Case Diagram for UC_12: Multi-Platform Access

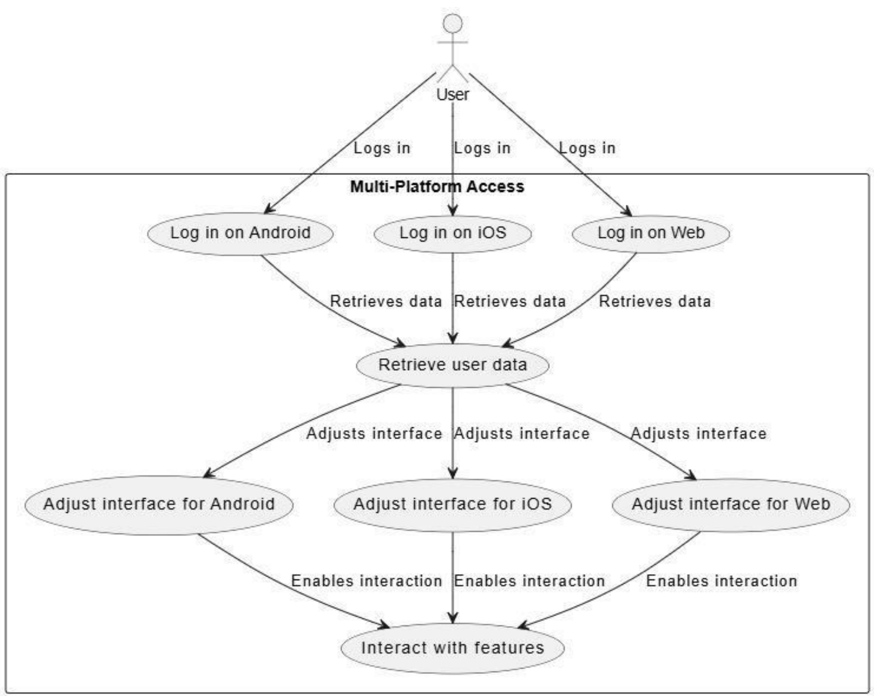


Figure 12: Use Case Diagram for UC_12: Multi-Platform Access

Table 15: Use Case Table for UC_12: Multi-Platform Access

Use Case ID	**UC_12**
Use Case Name	Multi-Platform Access
Description	Ensures users can access the system seamlessly across different platforms, such as Android, iOS, and web browsers.
Primary Actor	User
Secondary Actor	System
Pre-Condition	User account and data are synchronized across platforms.
Post- Condition	User experiences a consistent interface and functionality across platforms.
Basic Flow	1. User logs in from a platform (e.g., mobile app). 2. System retrieves user data and preferences. 3. Interface adjusts to the platform's requirements. 4. User interacts with the system seamlessly.
Alternate Flow	If a platform is unsupported, the system informs the user and provides alternatives.

3.1.13 Use Case Diagram for UC_13: User Feedback Submission

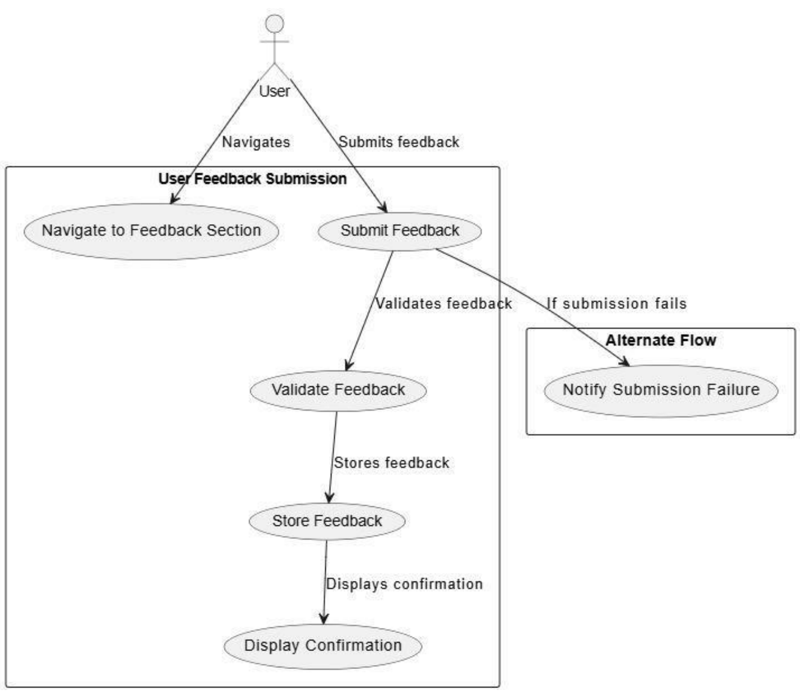


Figure 13: Use Case Diagram for UC_13: User Feedback Submission

Table 16: Use Case Table for UC_13: User Feedback Submission

Use Case ID	**UC_13**
Use Case Name	User Feedback Submission
Description	Allows users to submit feedback about their experience or report issues to improve the system.
Primary Actor	User
Secondary Actor	System
Pre-Condition	Feedback submission module is active, and the user is authenticated.
Post-Condition	Feedback is saved in the database for review by administrators.
Basic Flow	1. User navigates to the feedback section. 2. User submits feedback via a form (e.g., rating, comments). 3. System validates and stores the feedback. 4. Confirmation is displayed to the user.
Alternate Flow	If feedback submission fails, the system notifies the user and prompts them to retry.

3.2 Aggregated Use Case Diagram

The aggregated use case diagram represents all the system functionalities in a unified view. This diagram combines all individual use cases (voice cloning, avatar customization, real-time interaction, multilingual support, and cross-platform support) and shows how users interact with the system as a whole.

- **Actors**: The primary actor is the **User**, who interacts with the system across multiple functionalities. The **System** is the internal actor that processes data and responds to user inputs.
- **Use Cases**: The diagram includes all the use cases discussed earlier, such as **Voice Cloning**, **Avatar Customization**, **Real-Time Interaction**, **Multilingual Support**, and **Cross-Platform Support**.

3.2.1 Components of the Aggregated Use Case Diagram:

- **User**: Initiates all interactions with the system.
- **System**: Handles the processing and output for each functionality (e.g., voice cloning, avatar customization, multilingual support, etc.).
- **Use Cases**: Each individual functionality is represented as a use case in the system.

In the diagram:

- The system must handle each of these functionalities independently but ensure that they work together seamlessly, providing a smooth user experience.

3.2.2 Detailed Explanation

Each of these sections provides a detailed exploration of the system's use case analysis, from individual functionalities to an aggregated view of how the system supports complex, cross- platform operations. By breaking down the use case diagrams and description tables, we better understand the actors' roles and workflows that enable the system to meet the functional and non-functional requirements set in the earlier chapters.

This chapter ensures clarity on the interactions within the system, facilitating the development and integration of features that enhance user engagement, interaction, and satisfaction.

CHAPTER 4: ARCHITECTURAL AND DIAGRAMMATRIC REPRESENTATION

4.1 Architecture Diagram

The architecture diagram provides a high-level overview of the system, illustrating the interaction between its components. It highlights the relationships between the frontend (Flutter), backend (Firebase), and AI modules (TensorFlow and Unity).

- **Frontend (Flutter)**: Manages user interaction, providing a cross-platform interface for mobile and web users.
  - Handles UI components like forms, buttons, and avatars.
  - Communicates with the backend via REST APIs and Firebase SDKs.
- **Backend (Firebase)**: Manages data storage, user authentication, and cloud functions.
  - Real-time Database: Stores user data, cloned voices, and interaction logs.
  - Authentication: Handles secure user sign-ups and logins.
  - Cloud Functions: Executes server-side logic for voice processing and avatar customization.
- **AI Modules**:
  - **TensorFlow**: Implements voice cloning by processing user audio inputs and generating cloned voices.
  - **Unity**: Handles avatar customization with 3D rendering capabilities, allowing users to create personalized avatars.

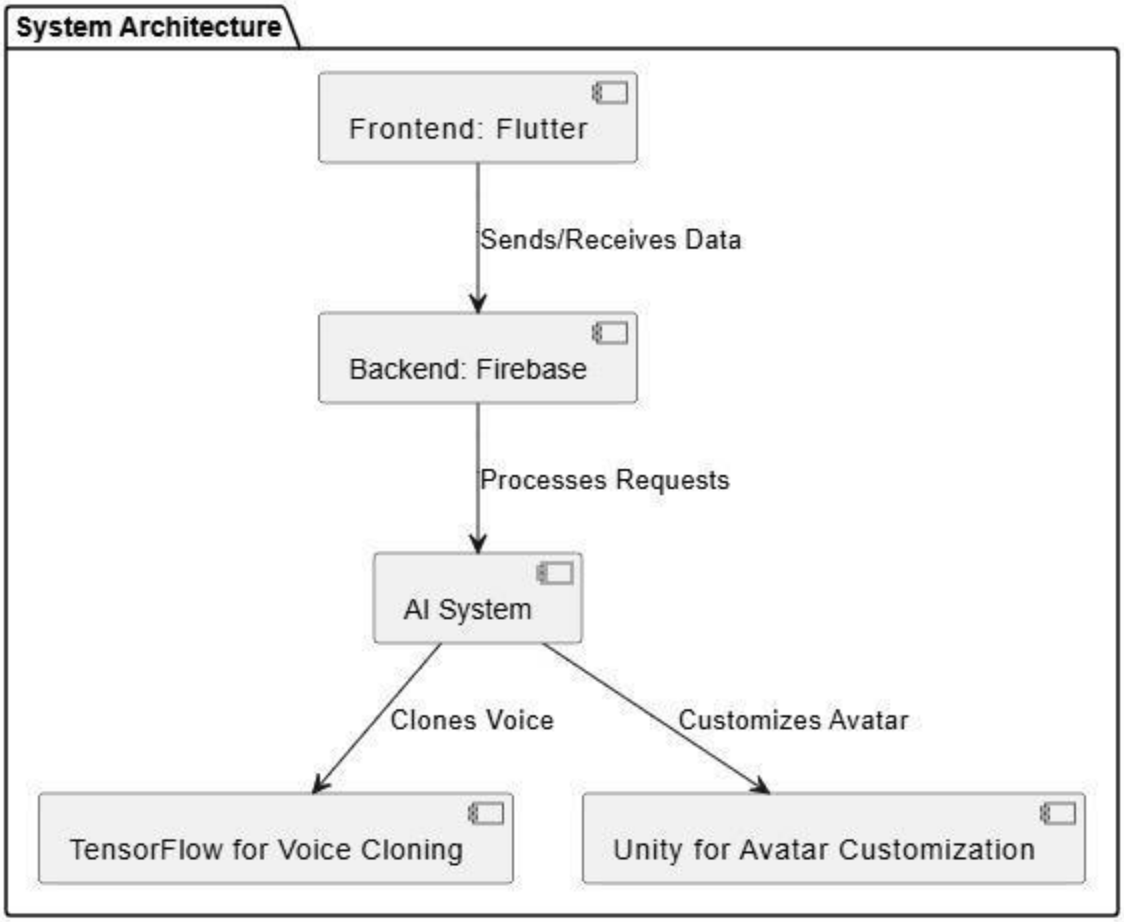


Figure 14.1: Architecture Diagram

4.2 Entity-Relationship Diagram (ERD) with Data Dictionary

The ERD depicts the relationships between key entities in the system. It ensures a clear understanding of how data flows and is stored.

- **Entities and Attributes**:
  - **User**: user_id, username, email, password_hash, preferences.
  - **VoiceData**: voice_id, user_id, audio_sample, cloned_voice.
  - **AvatarSettings**: avatar_id, user_id, customizations.
  - **InteractionLog**: interaction_id, user_id, input_text, output_text, timestamp.
- **Relationships**:
  - A Usercan have multiple VoiceDataand AvatarSettings.
  - InteractionLogtracks user interactions with the system.

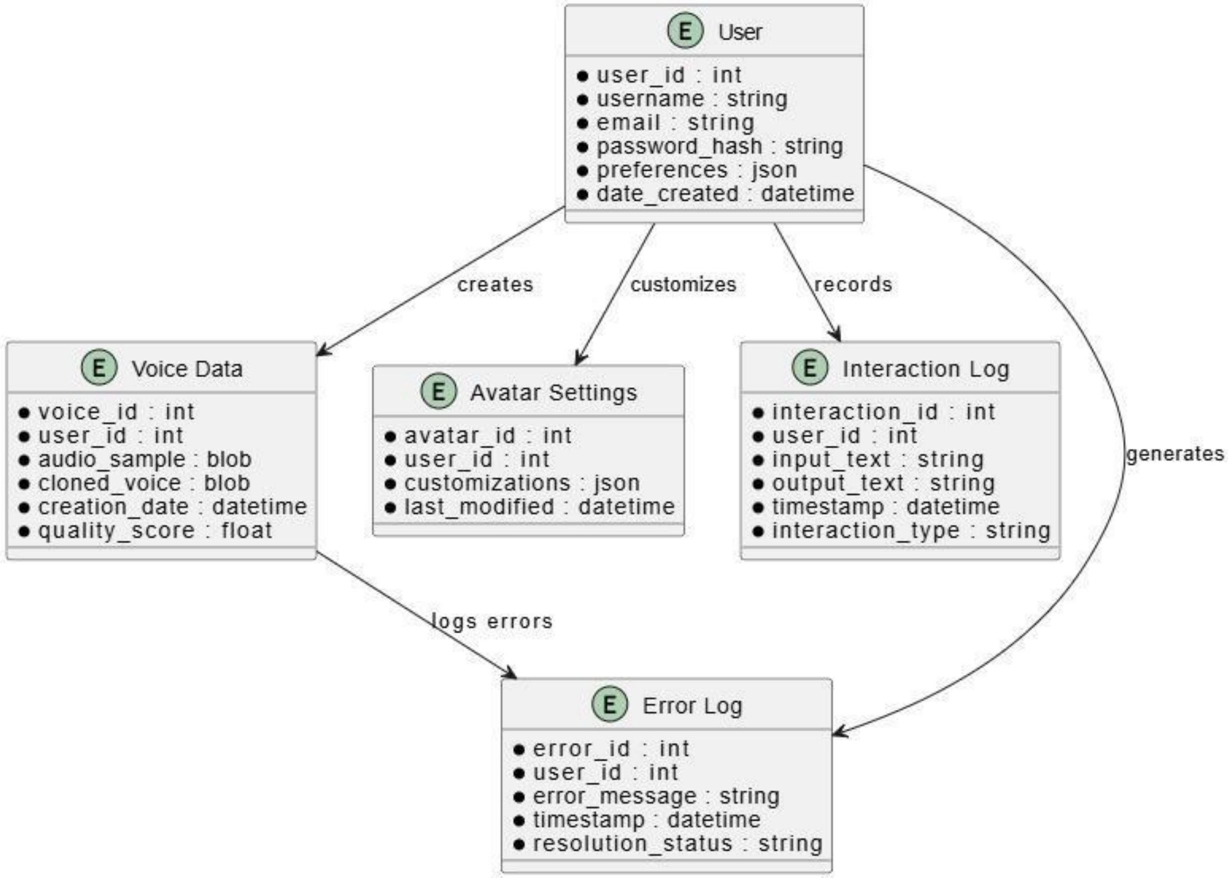


Figure 15: ERD Diagram

4.3 Data Flow Diagrams (DFD)

The DFDs represent the flow of data within and outside the system.

- **Level 0**: Shows the overall flow of information within the system, such as user inputs (voice/text), processing by AI modules, and responses.
- **Level 1**: Breaks down the flow into subsystems like voice cloning, avatar customization, and user authentication.

DFD Level 0:

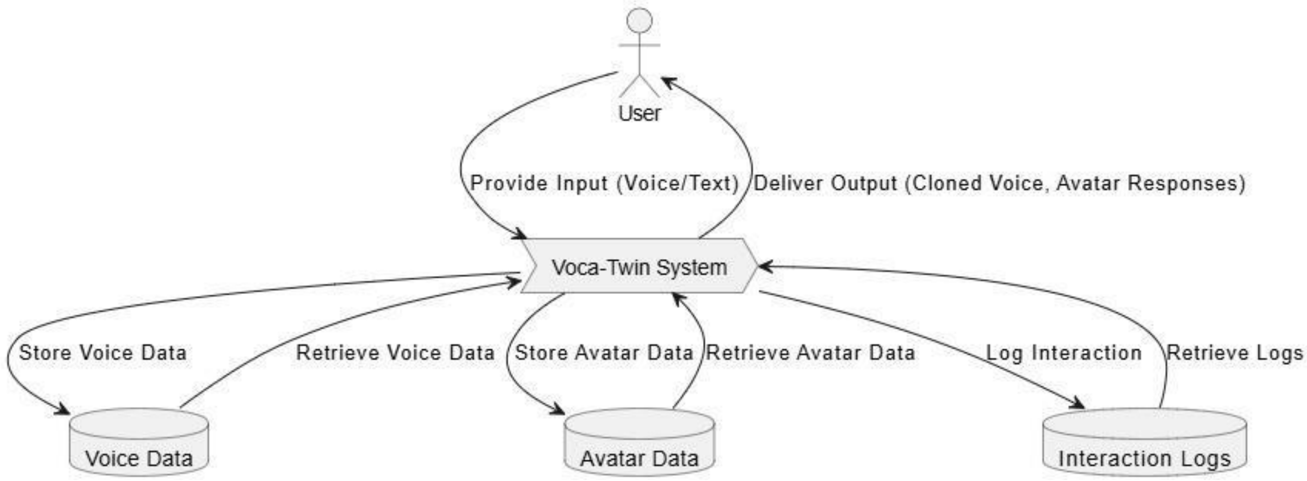


Figure 16: DFD Level 0

DFD Level 1:

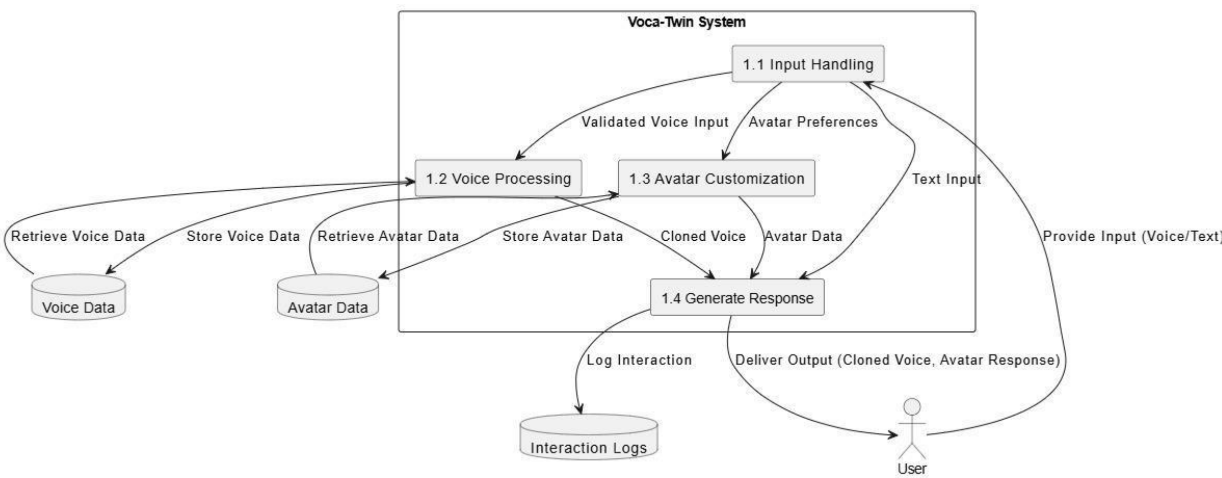


Figure 17: DFD Level 1

4.4 Class Diagram

The class diagram illustrates the system's structure, showing the attributes and methods of its classes.

- **User Class**: Manages user authentication and preferences.
- **VoiceData Class**: Handles voice recording, processing, and cloning.
- **AvatarSettings Class**: Manages avatar customization and rendering.
- **InteractionLog Class**: Tracks user interactions for analytics.

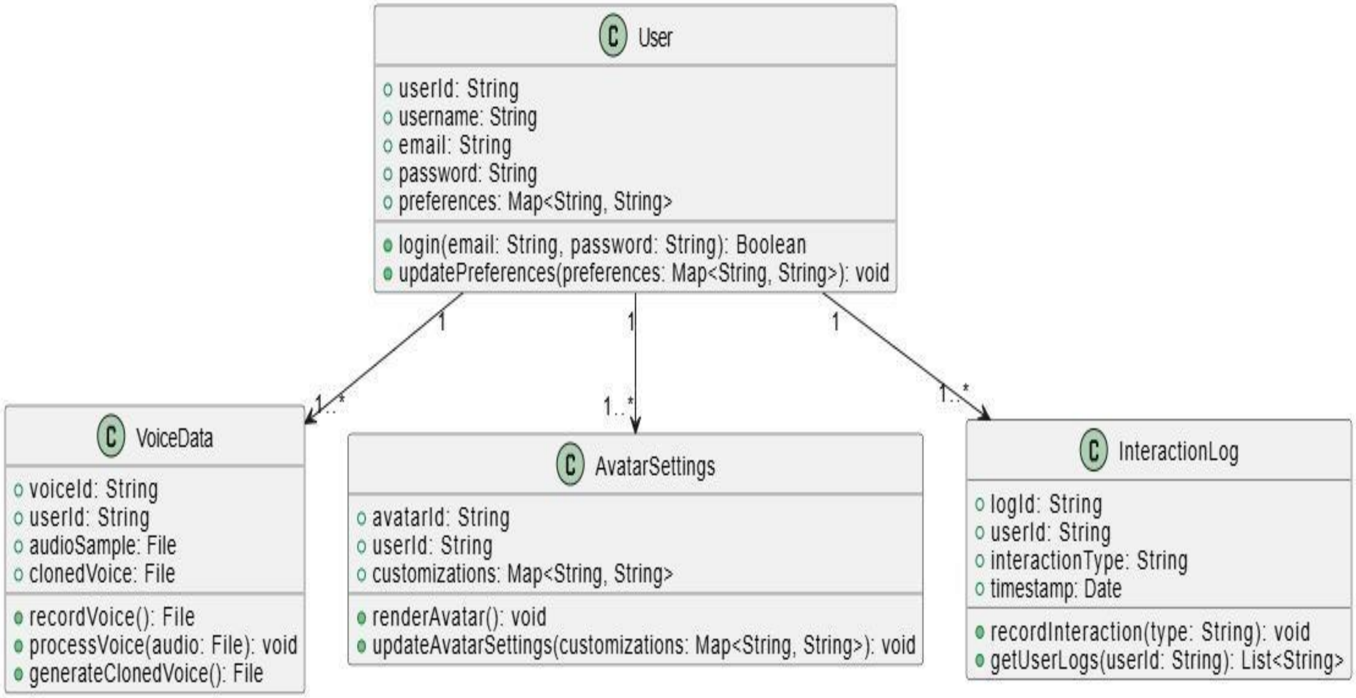


Figure 18: Class Diagram

4.5 Activity Diagrams

Activity diagrams depict the flow of actions for specific functionalities.

4.5.1 User Authentication

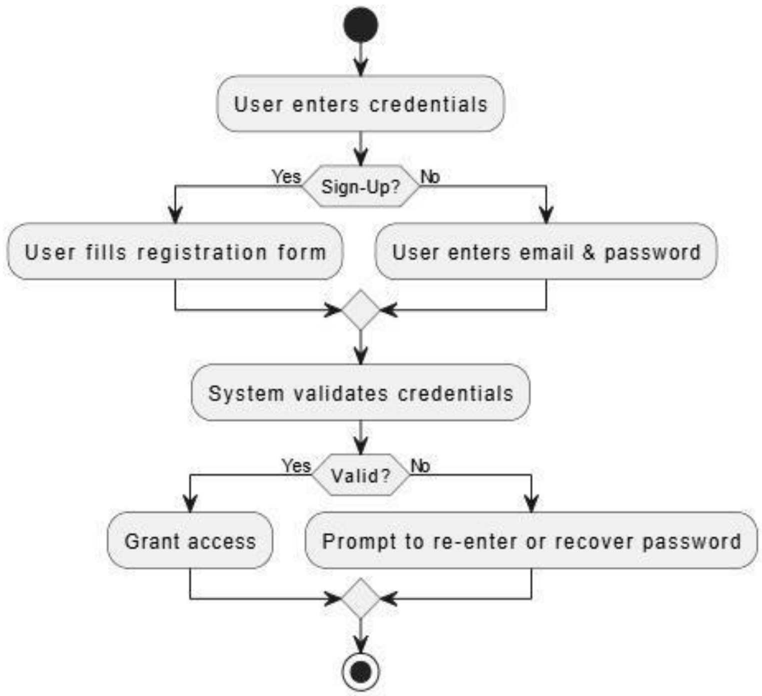


Figure 19: Activity Diagram for UC_01: User Authentication

4.5.2 Real time Voice cloning

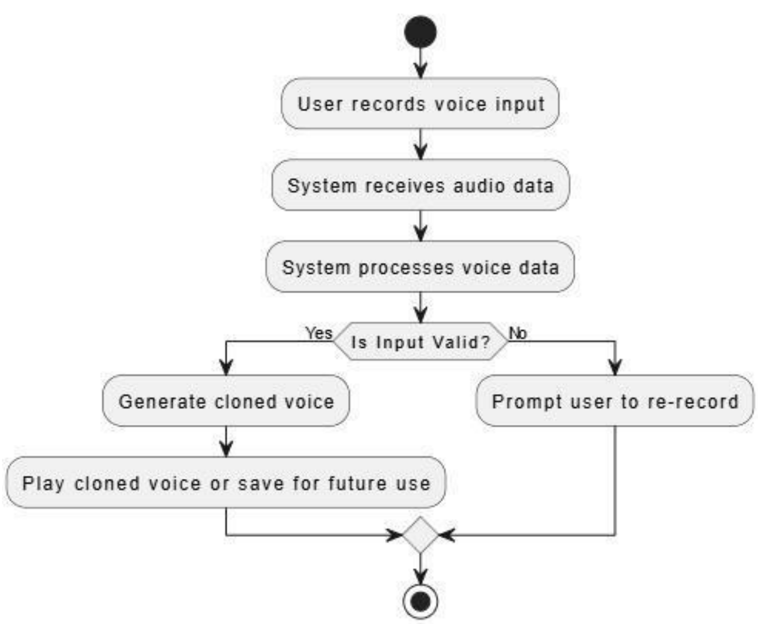


Figure 20: Activity Diagram for UC_02: Real-Time Voice Cloning

4.5.3 Real time Interaction

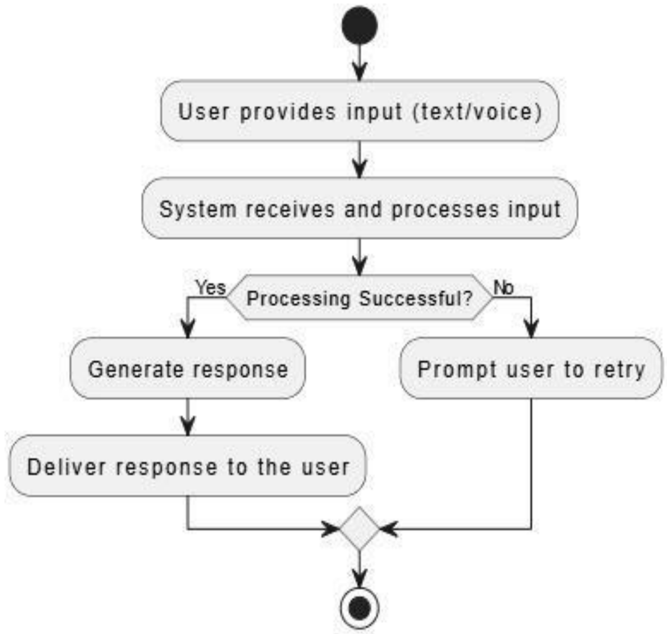


Figure 21: Activity Diagram for UC_03: Real-Time Interaction

4.5.4 Text to speech

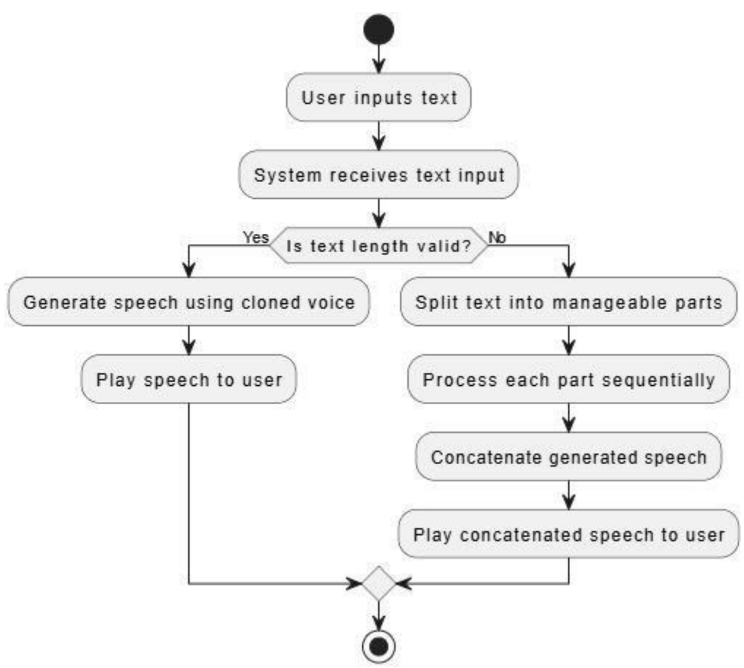


Figure 22: Activity Diagram for UC_04: Text to speech

4.5.5 AI Personalized Chatbot

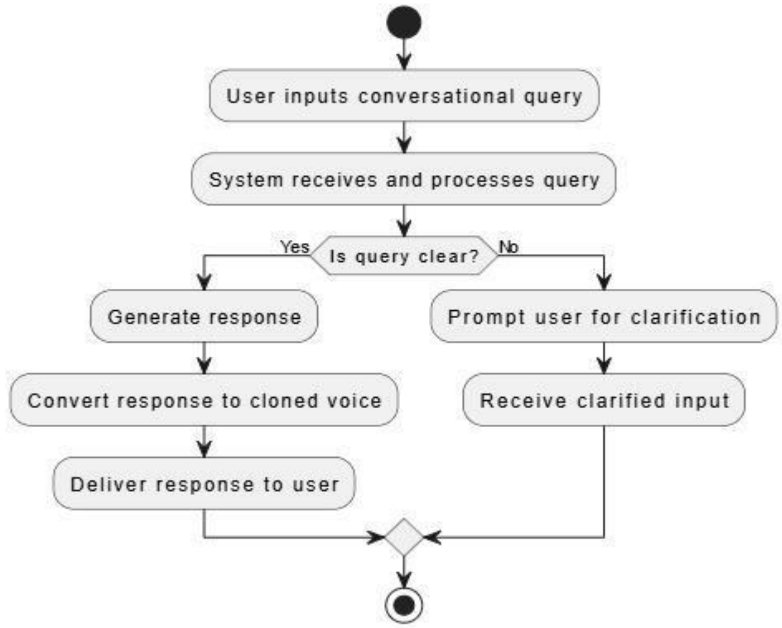


Figure 23: Activity Diagram for UC_13: AI Personalized Chatbot

4.5.6 Accessability of user interaction

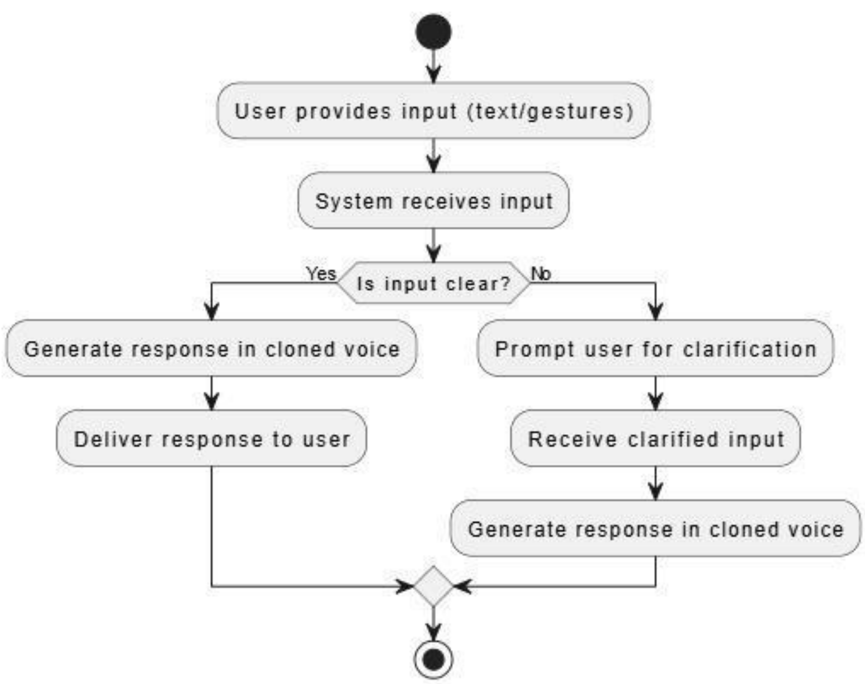


Figure 24: Activity Diagram for UC_06: Accessability of User Interaction

4.5.7 Multilingual Support

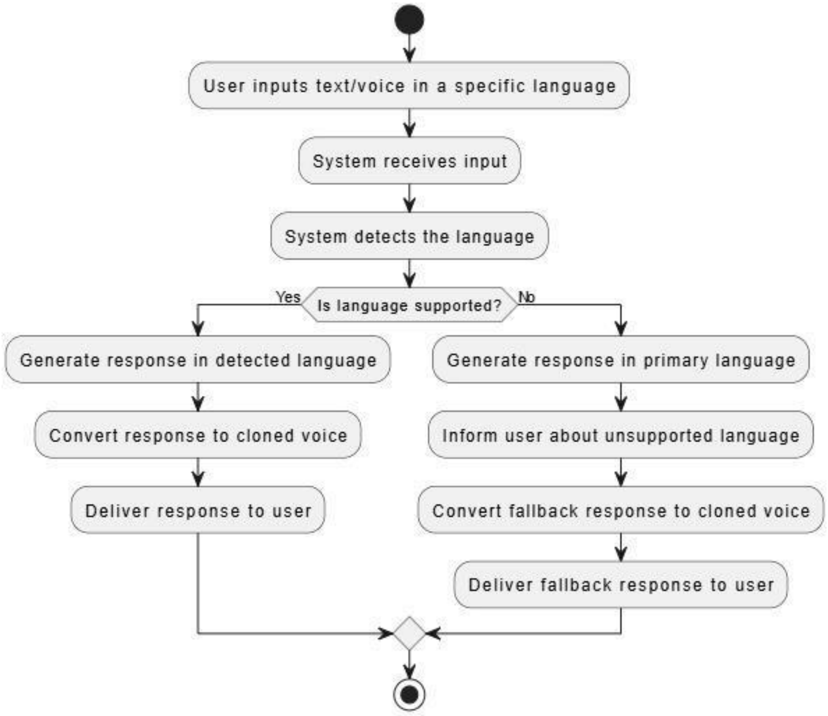


Figure 25: Activity Diagram for UC_07: Multilingual Support

4.5.8 Virtual Meeting

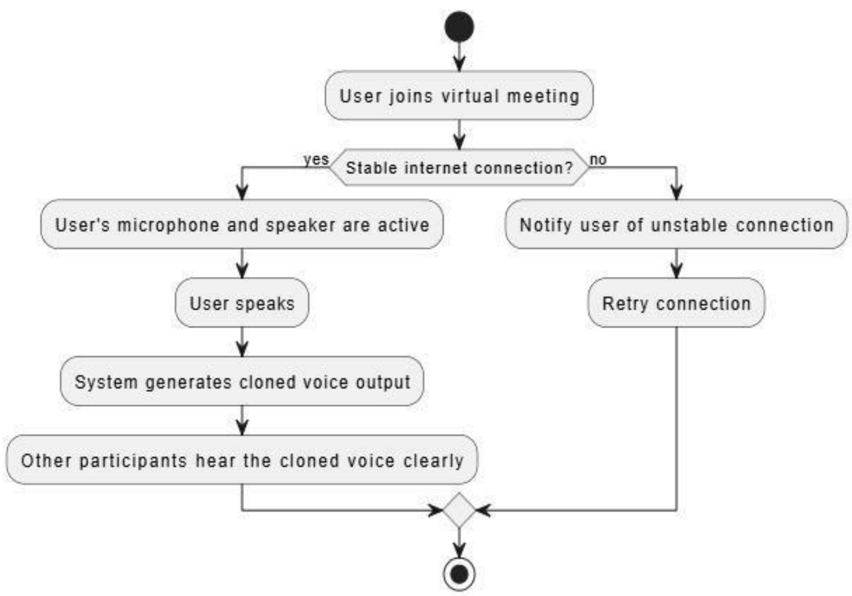


Figure 26: Activity Diagram for UC_08: Virtual Meeting

4.5.9 Remainder

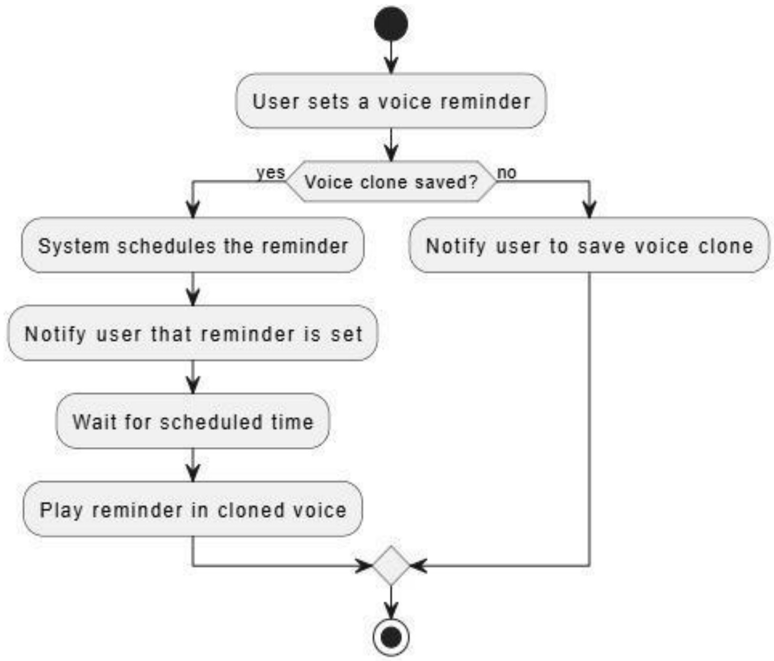


Figure 27: Activity Diagram for UC_09: Remainder

4.5.10 Avatar Creation and Personalization

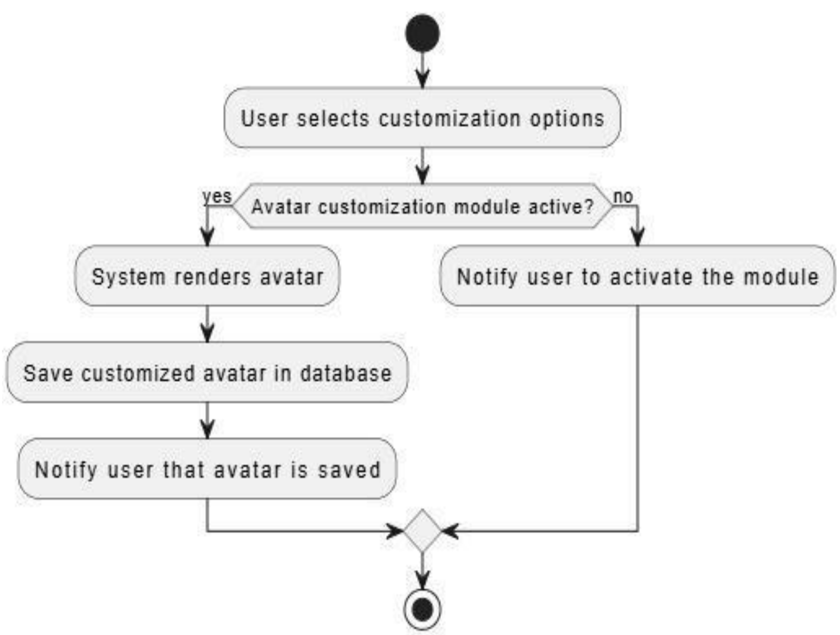


Figure 28: Activity Diagram for UC_10: Avatar Creation and Personalization

4.5.11 Interactive voice-based customer support

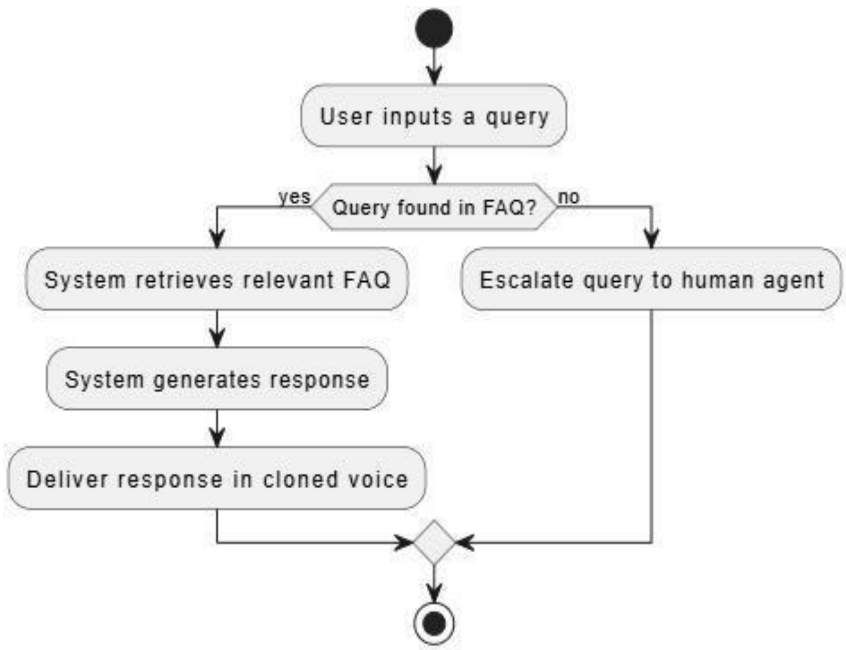


Figure 29: Activity Diagram for UC_11: Interactive voice-based customer support

4.5.12 Multi-platform Access

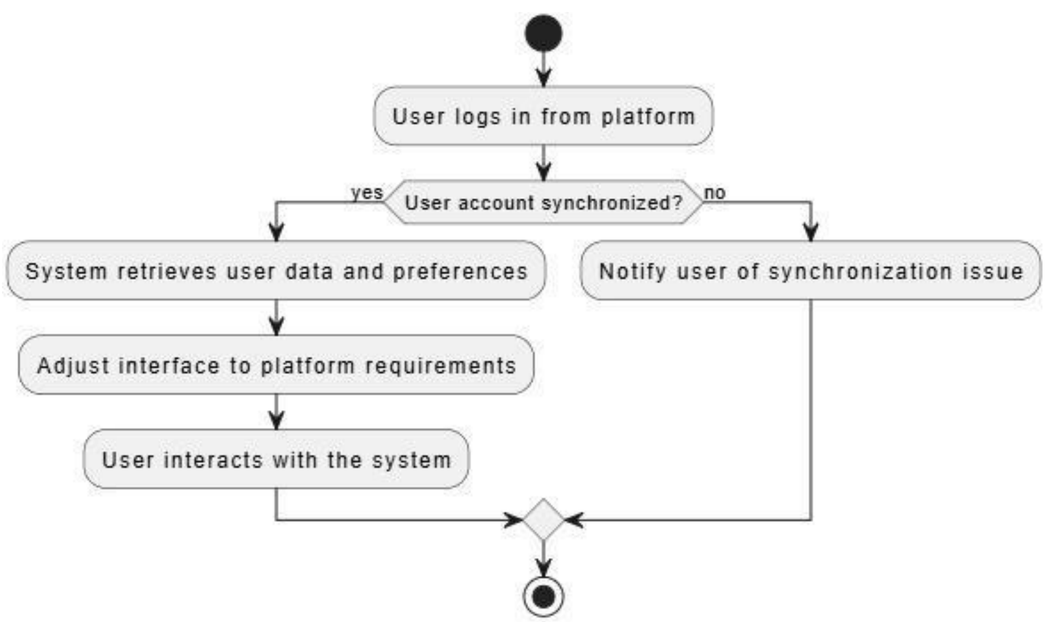


Figure 30: Activity Diagram for UC_12: Multi-Platform Access

4.5.13 User feedback

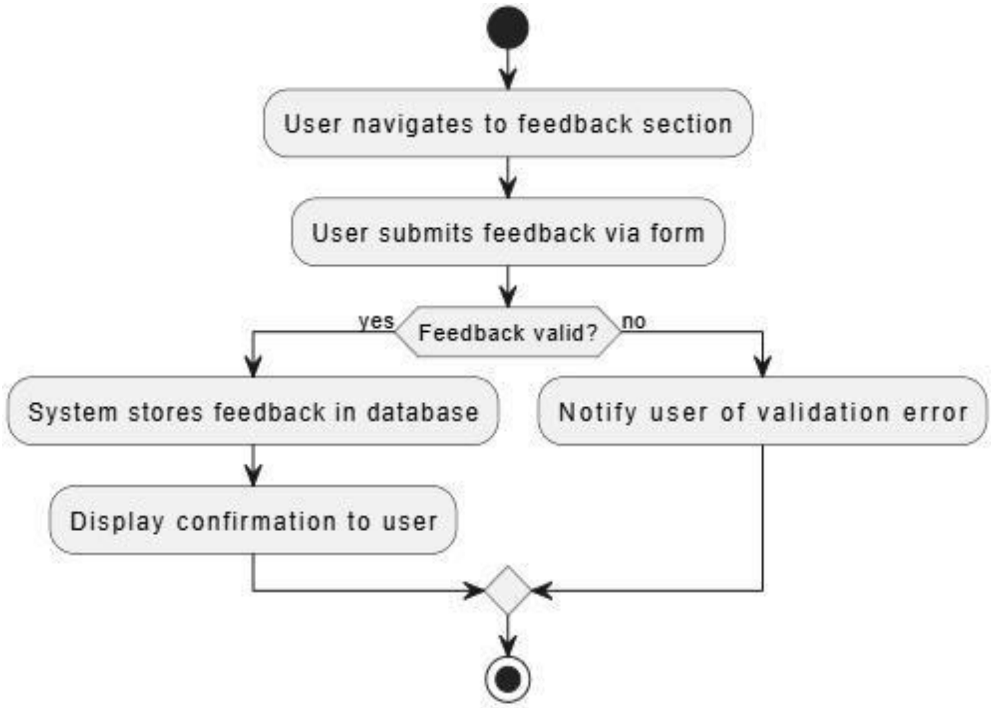


Figure 31: Activity Diagram for UC_13: User Feedback

4.6 Sequence Diagrams

Sequence diagrams illustrate the interaction between system components and the order of operations for specific functionalities.

4.6.1 User Authentication

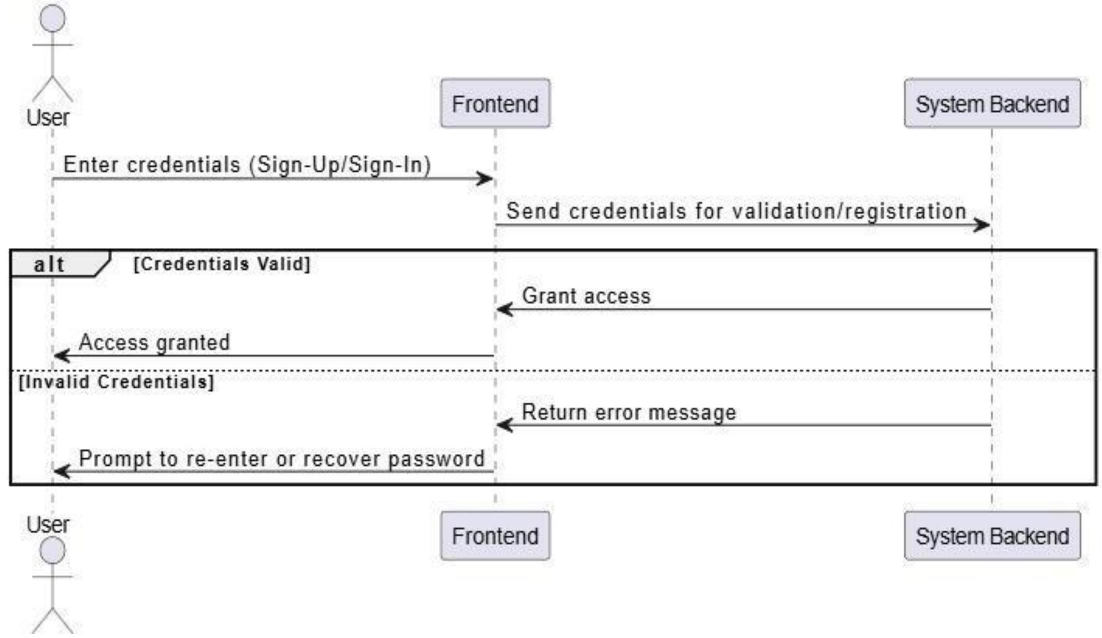


Figure 32: Sequence Diagram for UC_01: User Authentication

4.6.2 Real time Voice cloning

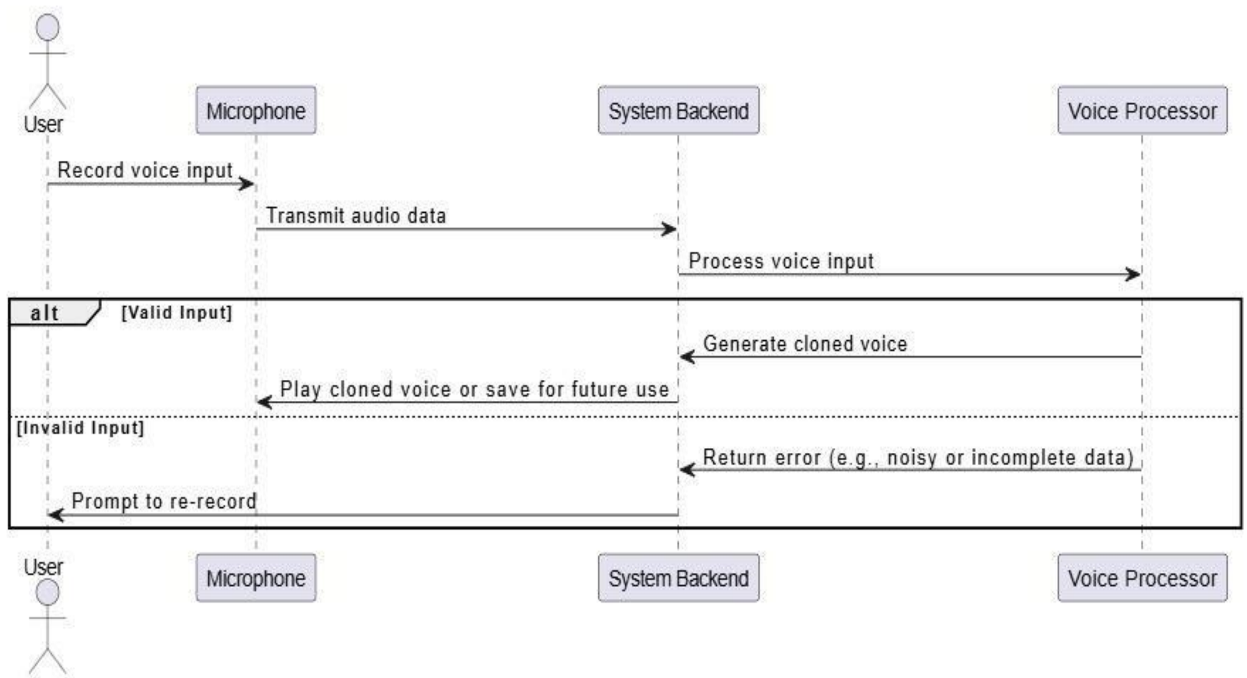


Figure 33: Sequence Diagram for UC_02: Real-time Voice Cloning

4.6.3 Real time Interaction

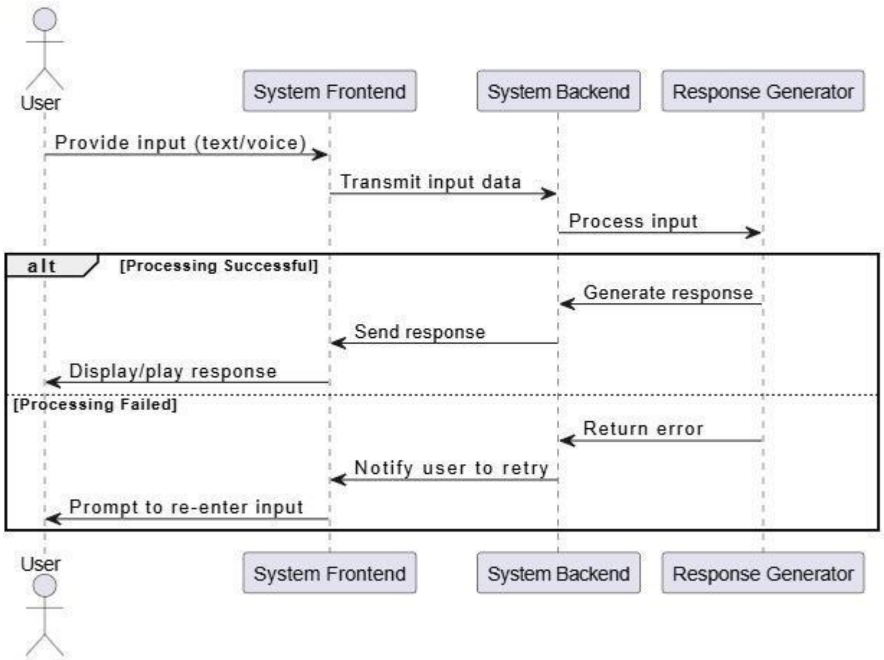


Figure 34: Sequence Diagram for UC_03: Real-Time Interaction

4.6.4 Text to speech

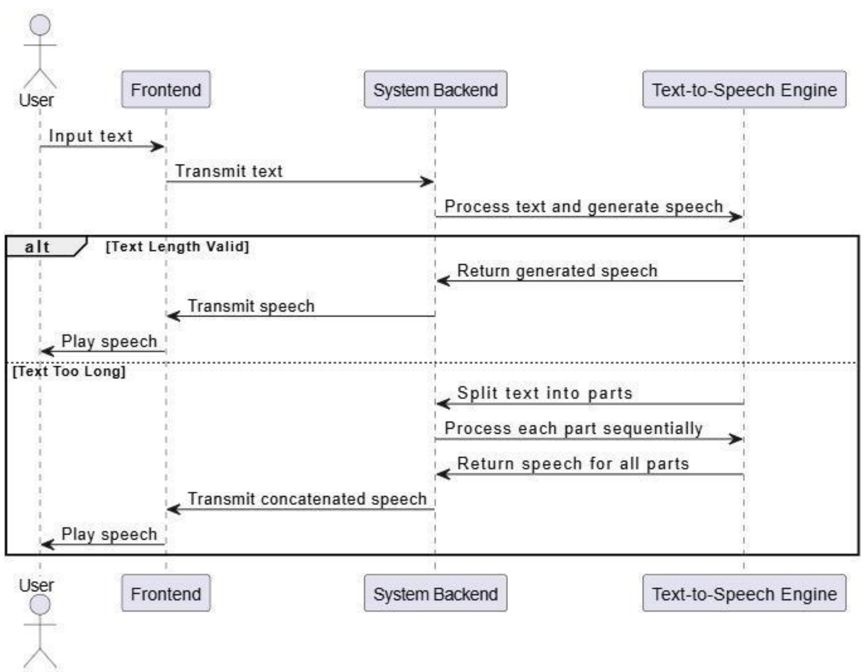


Figure 35: Sequence Diagram for UC_04: Text to Speech

4.6.5 AI Personalized Chatbot

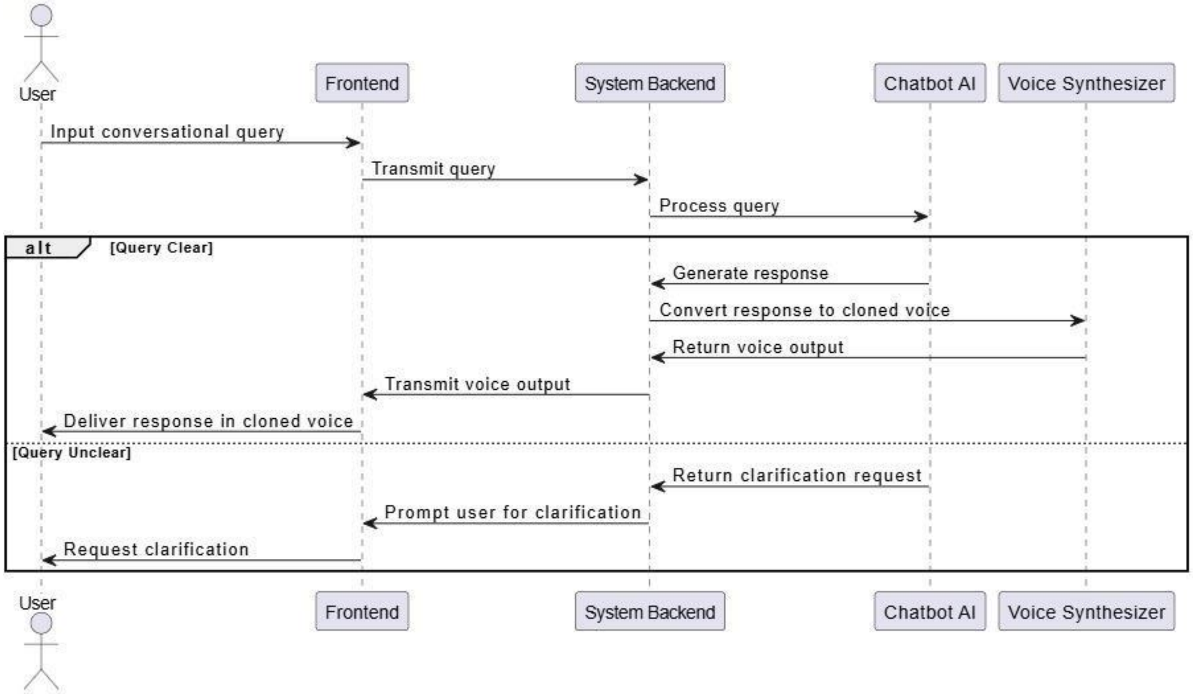


Figure 36: Sequence Diagram for UC_05: AI Personalized Chatbot

4.6.6 Accessability of user interaction

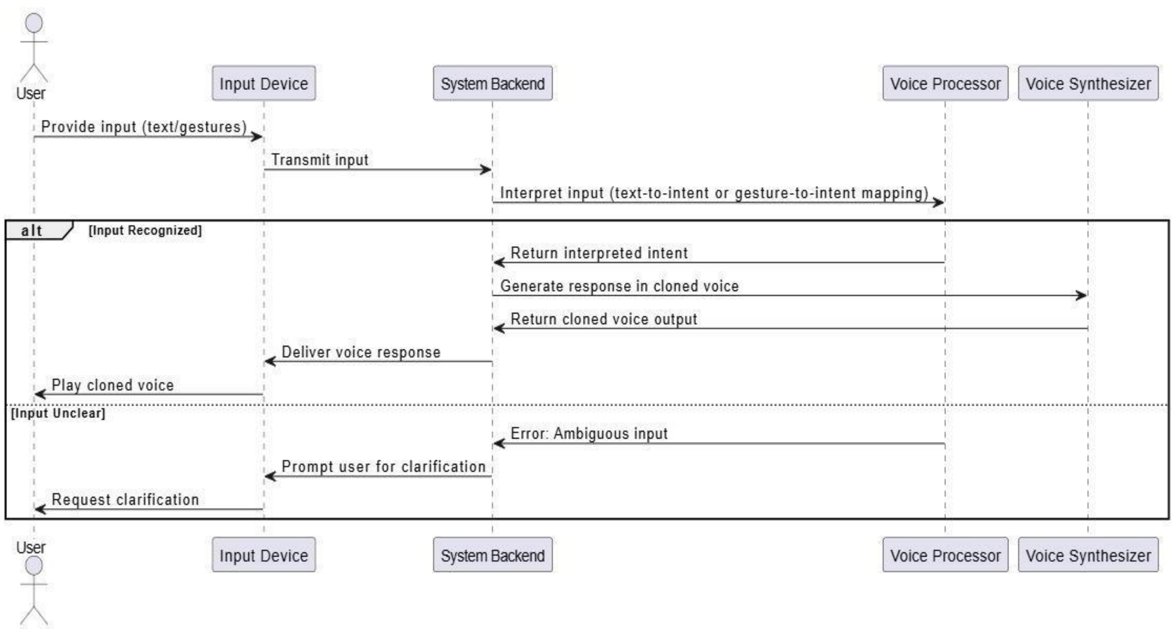


Figure 37: Sequence Diagram for UC_06: Accessability of User Interaction

4.6.7 Multilingual Support

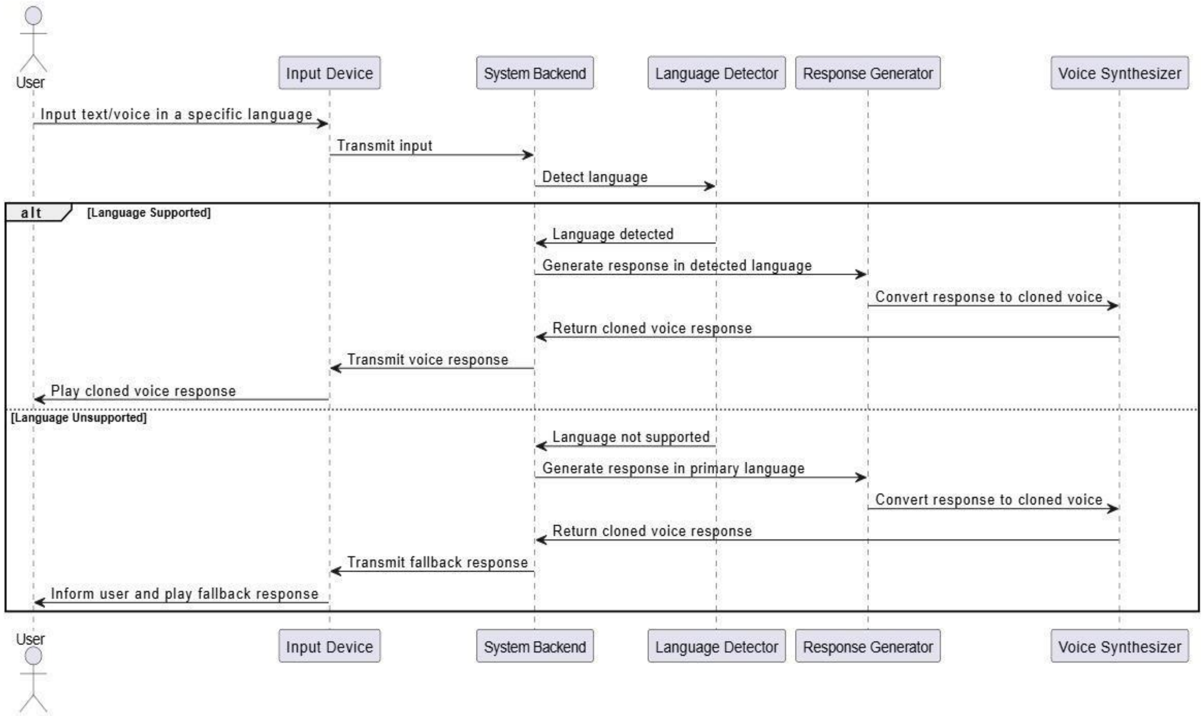


Figure 38: Sequence Diagram for UC_07: Multilingual Support

4.6.8 Virtual Meeting

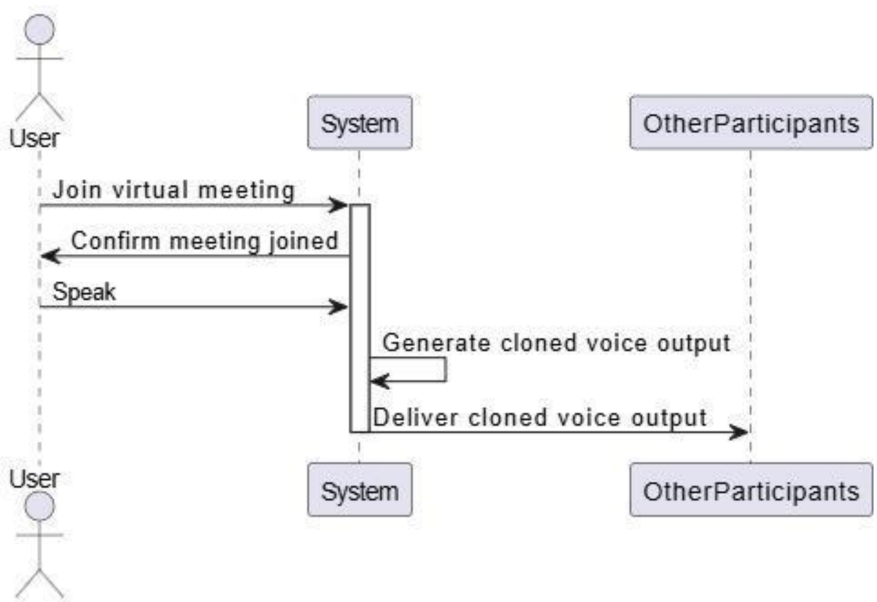


Figure 39: Sequence Diagram for UC_08: Virtual Meeting

4.6.9 Remainder

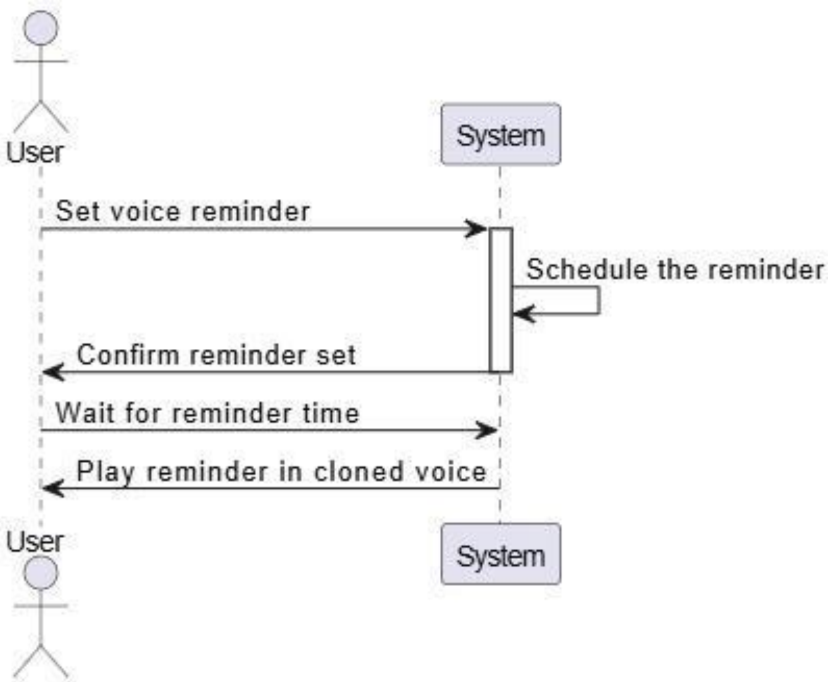


Figure 40: Sequence Diagram for UC_09: Remainder

4.6.10 Avatar Creation and Personalization

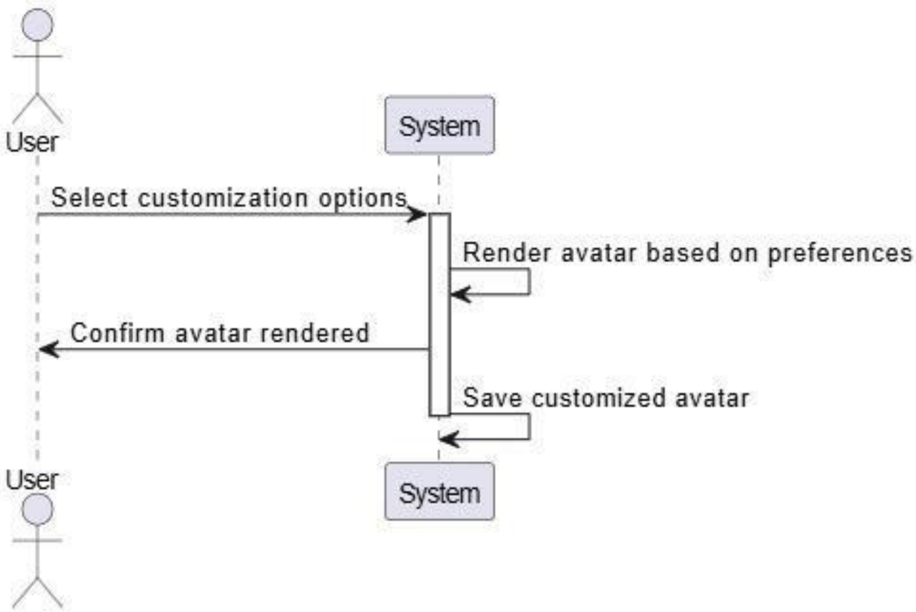


Figure 41: Sequence Diagram for UC_10: Avatar Creation and Personalization

4.6.11 Interactive voice-based customer support

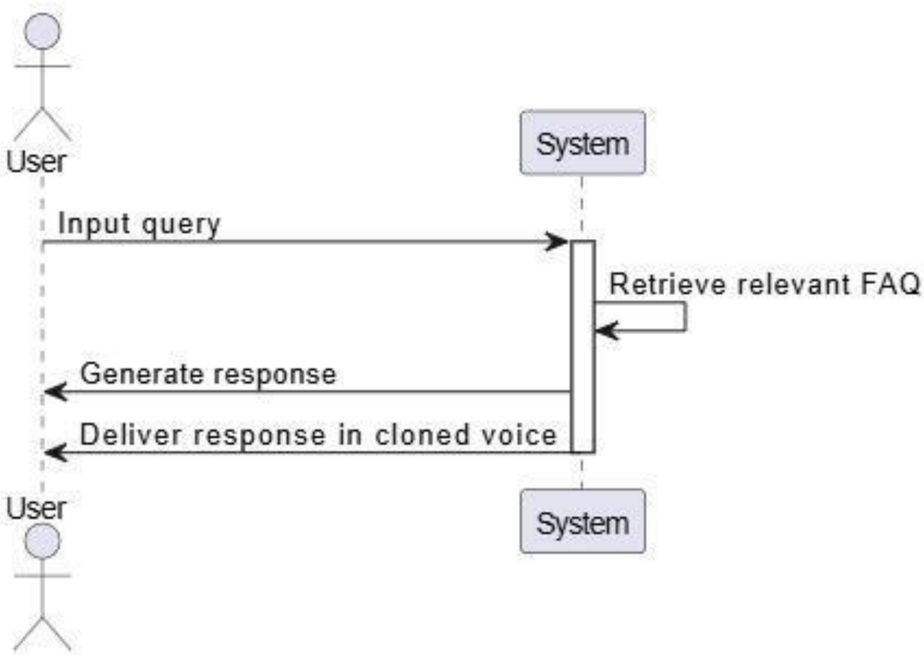


Figure 42: Sequence Diagram for UC_11: Interactive voice-based customer support

4.6.12 Multi-platform Access

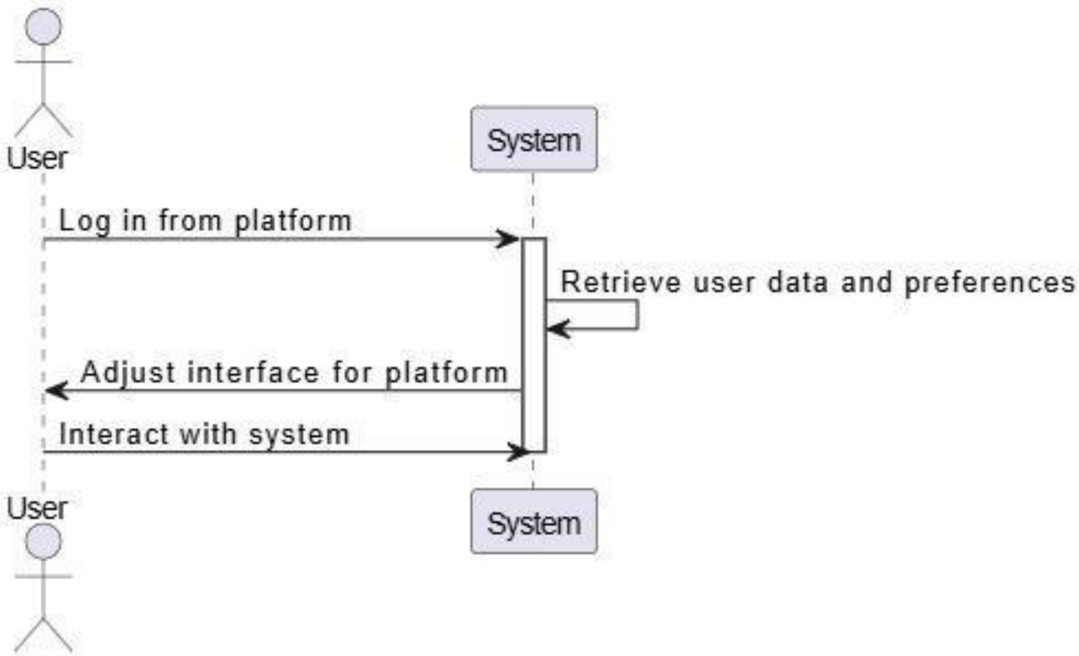


Figure 43: Sequence Diagram for UC_12: Multi-Platform Access

4.6.13 User feedback

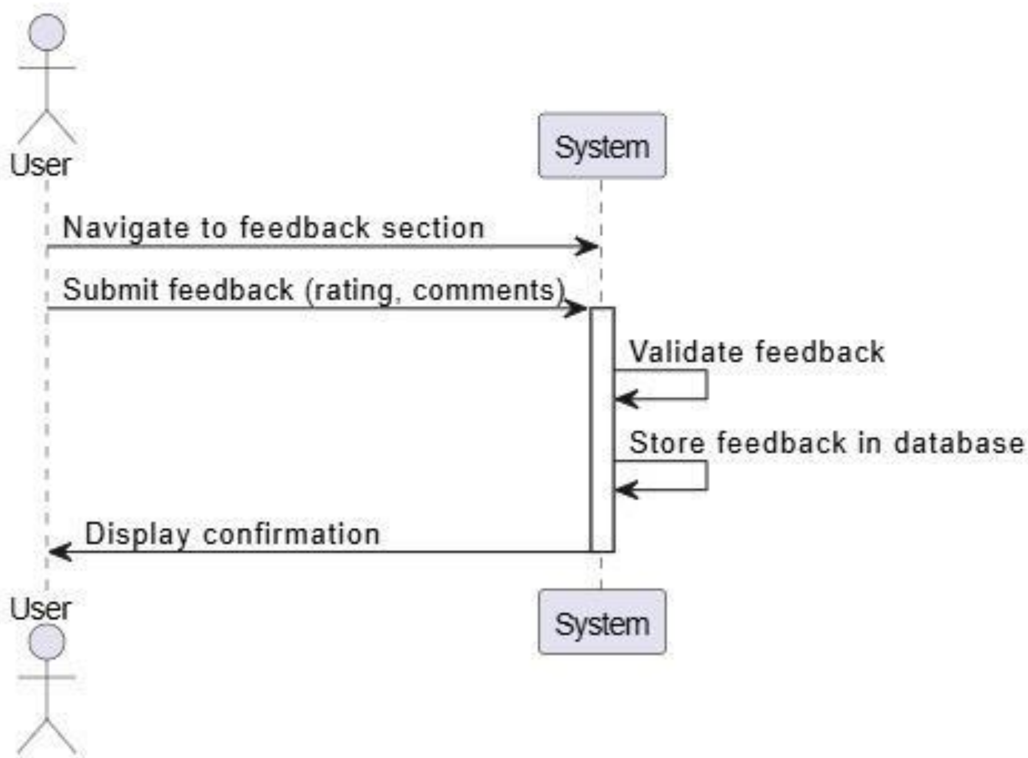


Figure 44: Sequence Diagram for UC_13: User Feedback

4.7 Collaboration Diagrams

Collaboration diagrams, also known as communication diagrams, depict the relationships and interactions between objects or components in the system. They focus on the structural organization of the system.

4.7.1 User Authentication

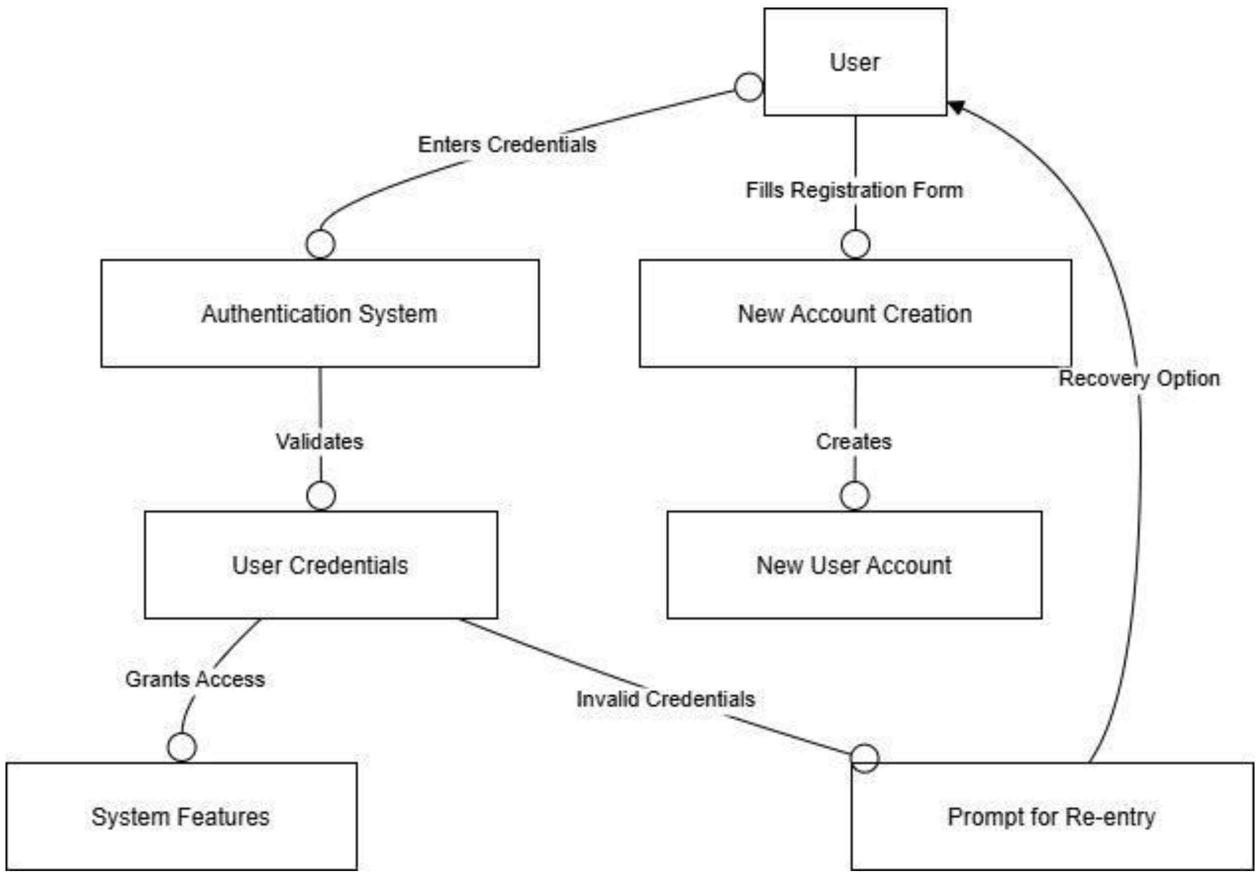


Figure 45: Collaboration Diagram for UC_01: User Authentication

4.7.2 Real time Voice cloning

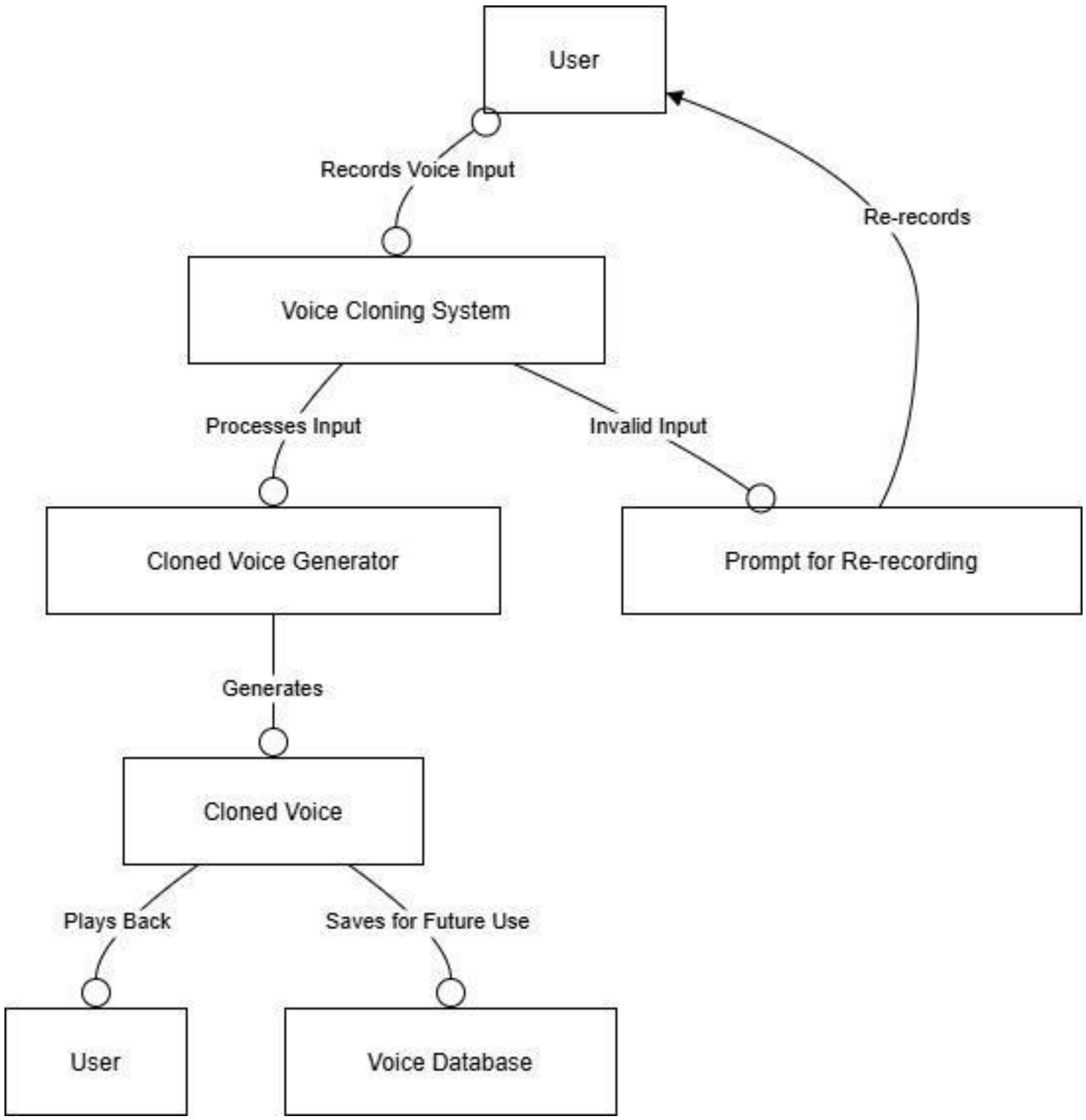


Figure 46: Collaboration Diagram for UC_02: Real-time Voice cloning

4.7.3 Real time Interaction

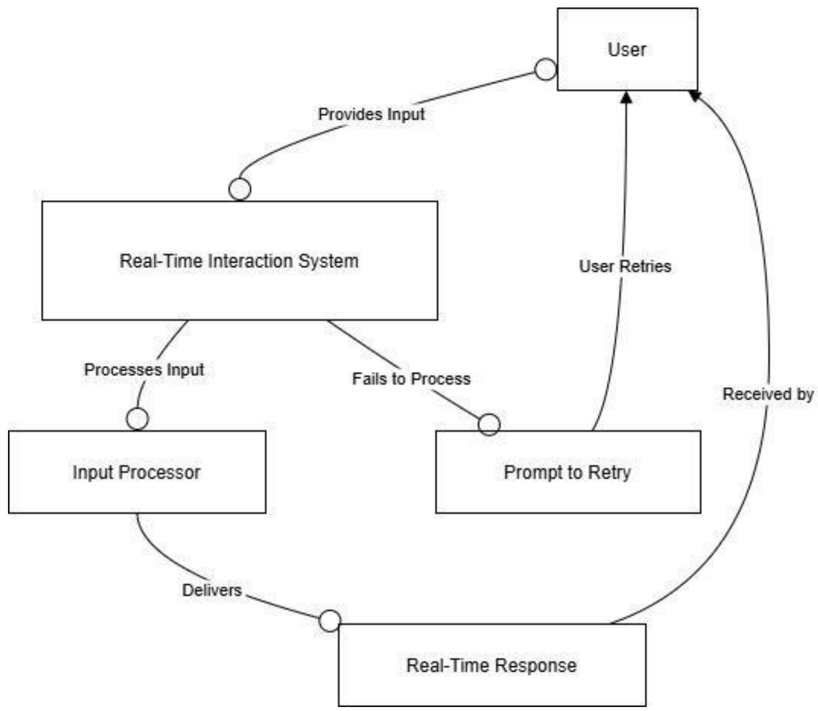


Figure 47: Collaboration Diagram for UC_03: Real-Time Interaction

4.7.4 Text to speech

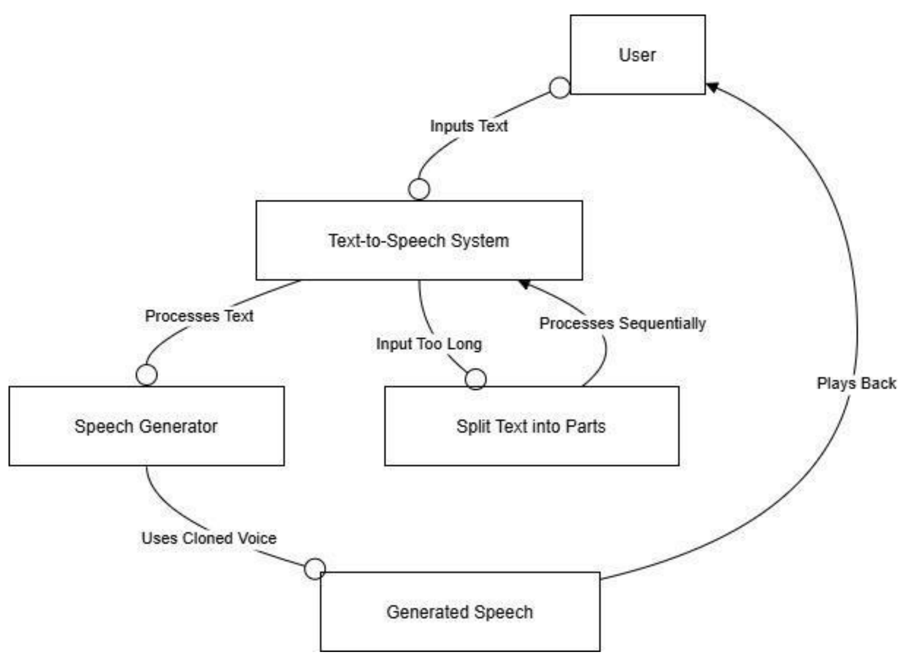


Figure 48: Collaboration Diagram for UC_04: Text to Speech

4.7.5 AI Personalized Chatbot

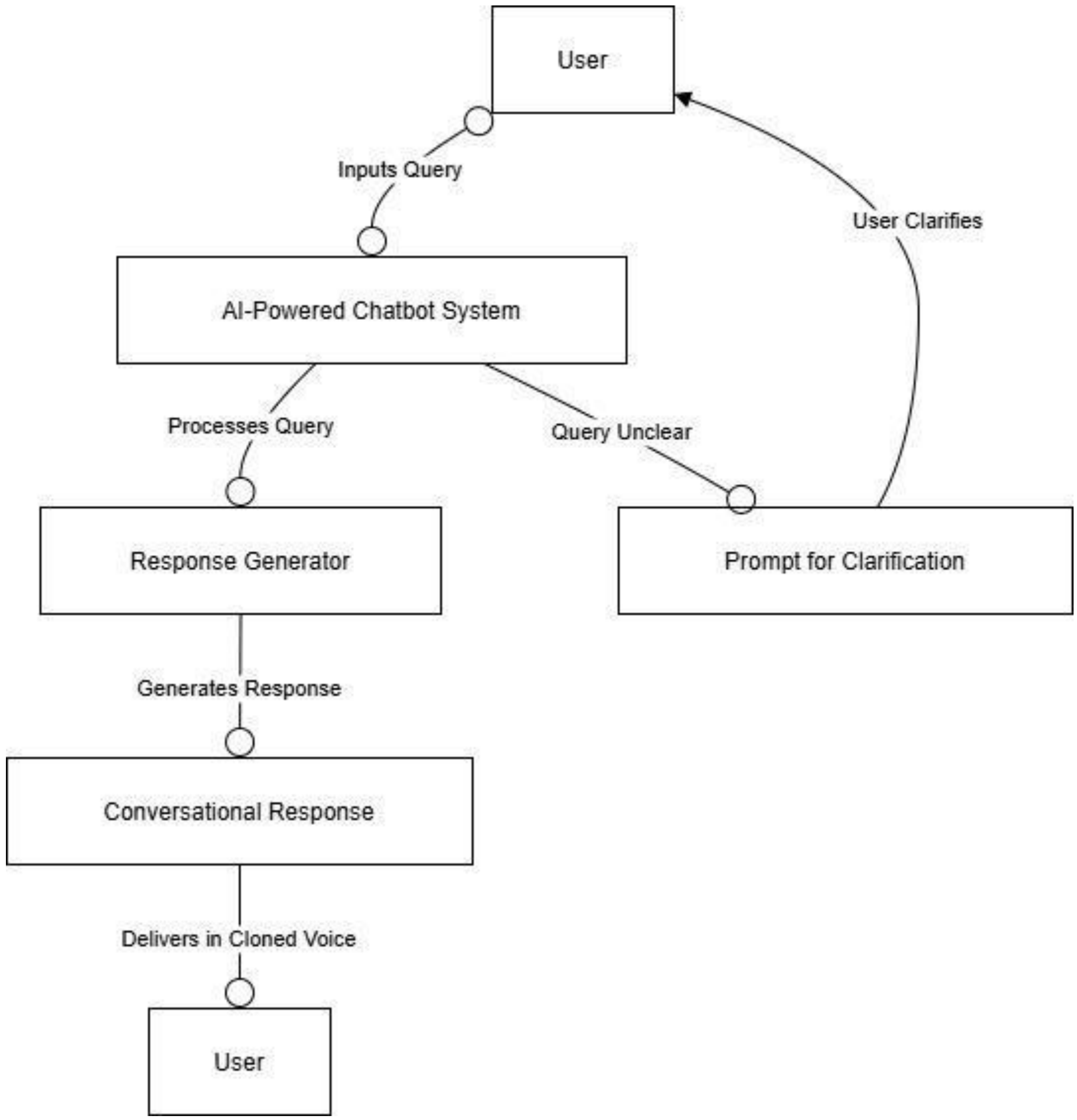


Figure 49: Collaboration Diagram for UC_05: AI Personalized Chatbot

4.7.6 Accessibility of user interaction

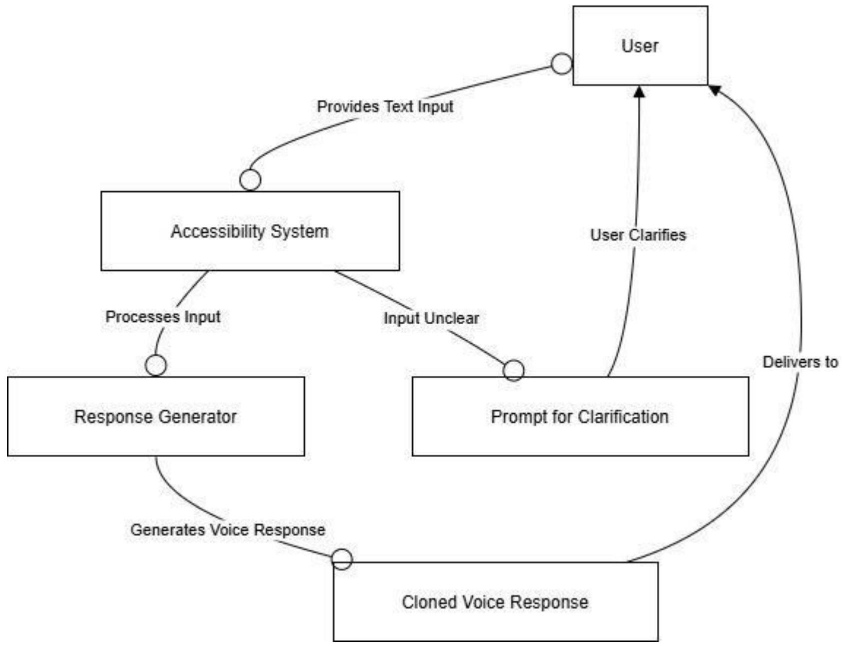


Figure 50: Collaboration Diagram for UC_06: Accessibility of User Interaction

4.7.7 Multilingual Support

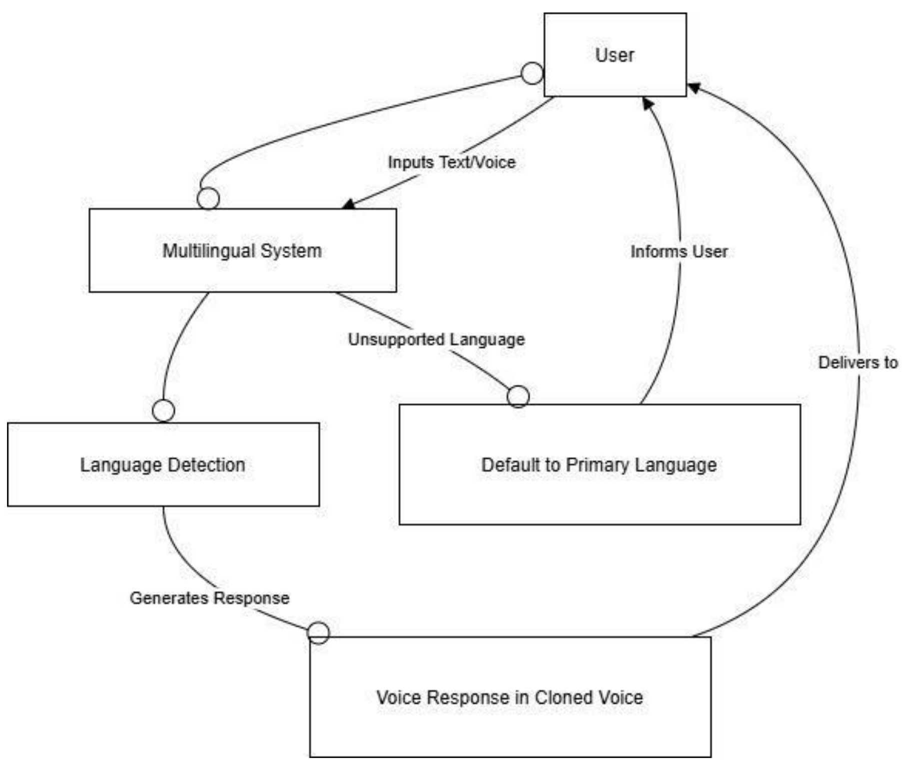


Figure 51: Collaboration Diagram for UC_07: Multilingual Support

4.7.8 Virtual Meeting

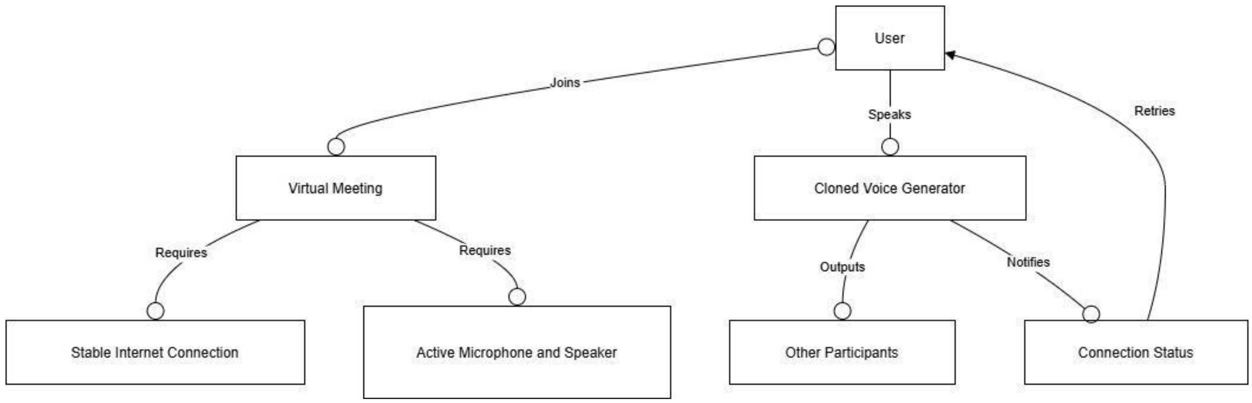


Figure 52: Collaboration Diagram for UC_08: Virtual Meeting

4.7.9 Remainder

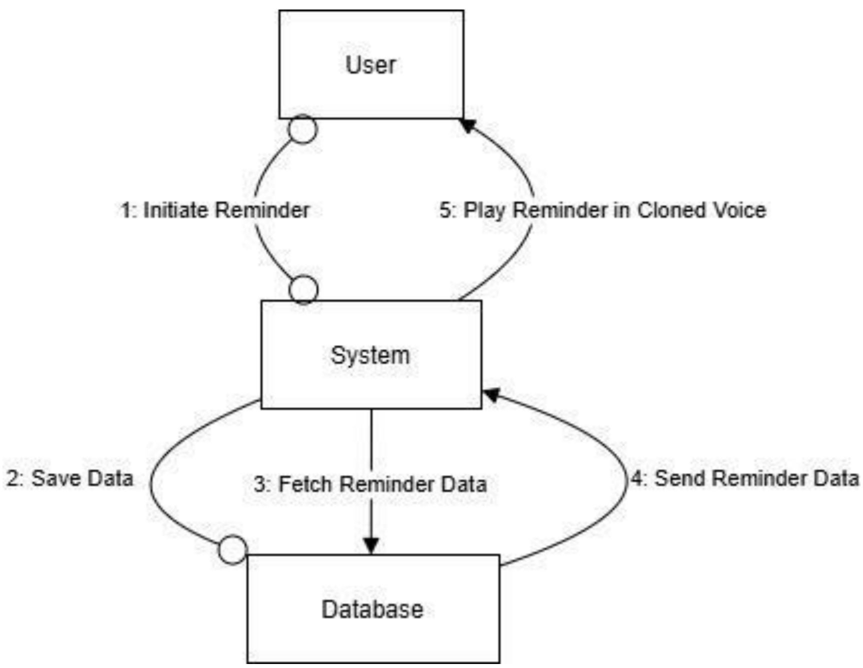


Figure 53: Collaboration Diagram for UC_09: Remainder

4.7.10 Avatar Creation and Personalization

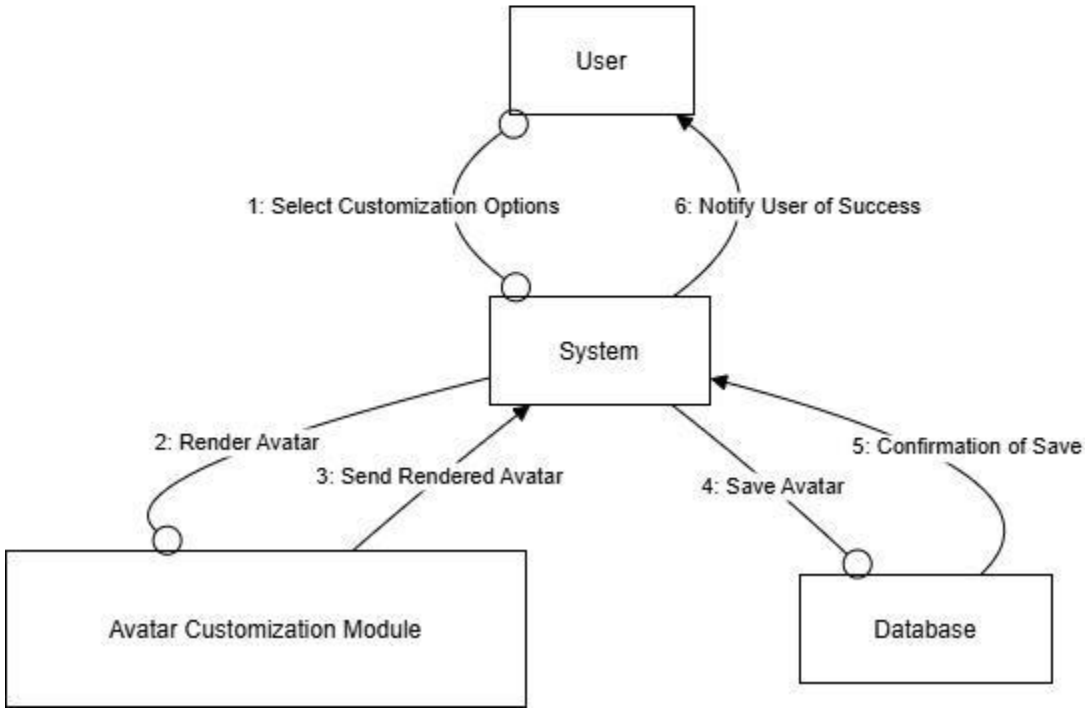


Figure 54: Collaboration Diagram for UC_10: Avatar Creation and Personalization

4.7.11 Interactive voice-based customer support

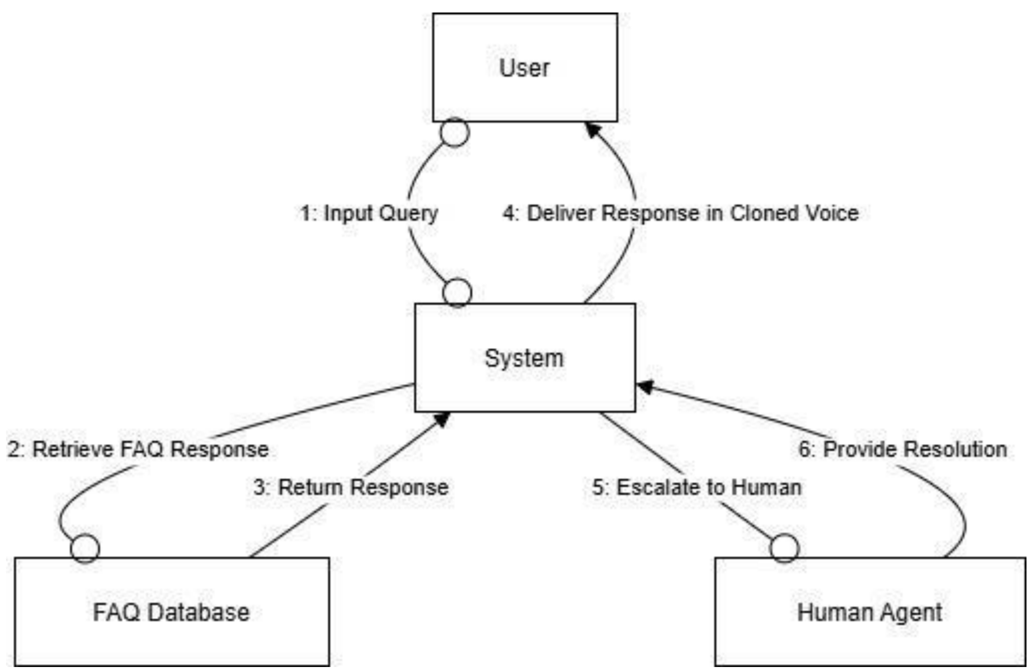


Figure 55: Collaboration Diagram for UC_11: Interactive voice-based customer support

4.7.12 Multi-platform Access

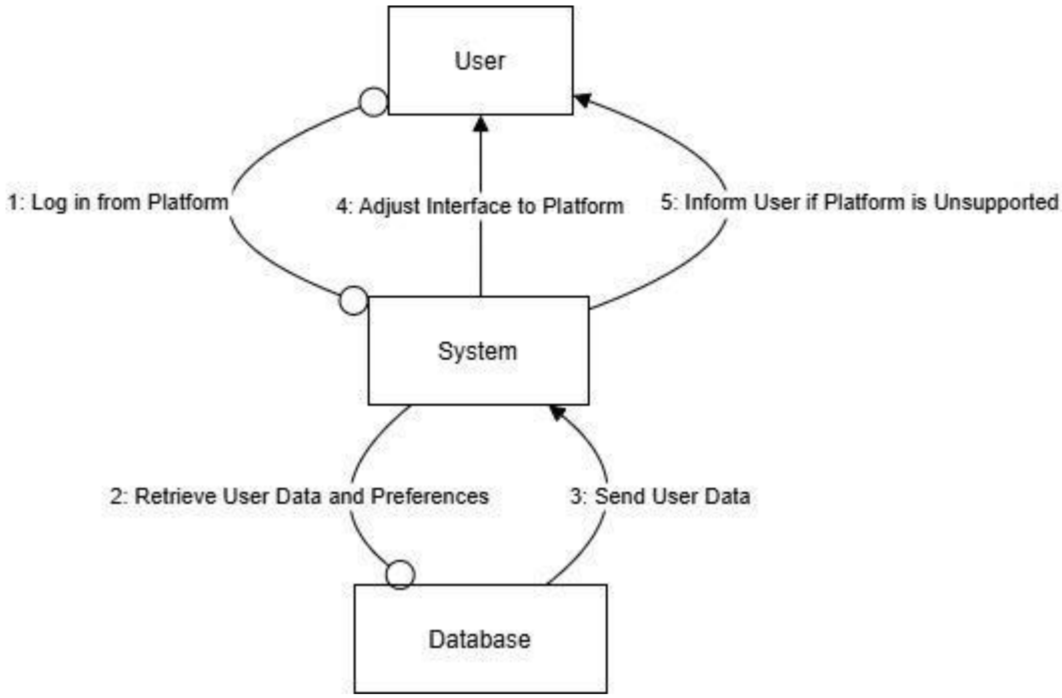


Figure 56: Collaboration Diagram for UC_12: Multi-Platform Access

4.7.13 User feedback

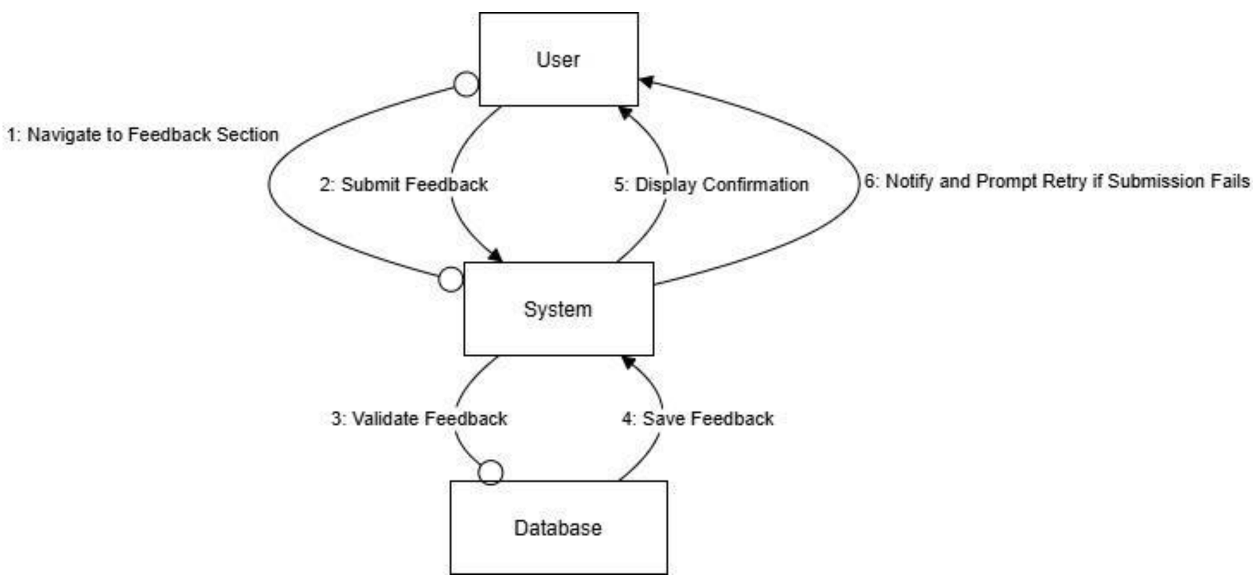


Figure 57: Collaboration Diagram for UC_13: User Feedback

4.8 State Transition Diagram

This diagram illustrates the states of the system and transitions triggered by user actions or system events.

- **States**:
  - Idle
  - Voice Recording
  - Voice Processing
  - Voice Cloning
  - Avatar Customization
- **Transitions**:
  - Triggered by user inputs or system responses, such as starting/stopping a process or saving data.

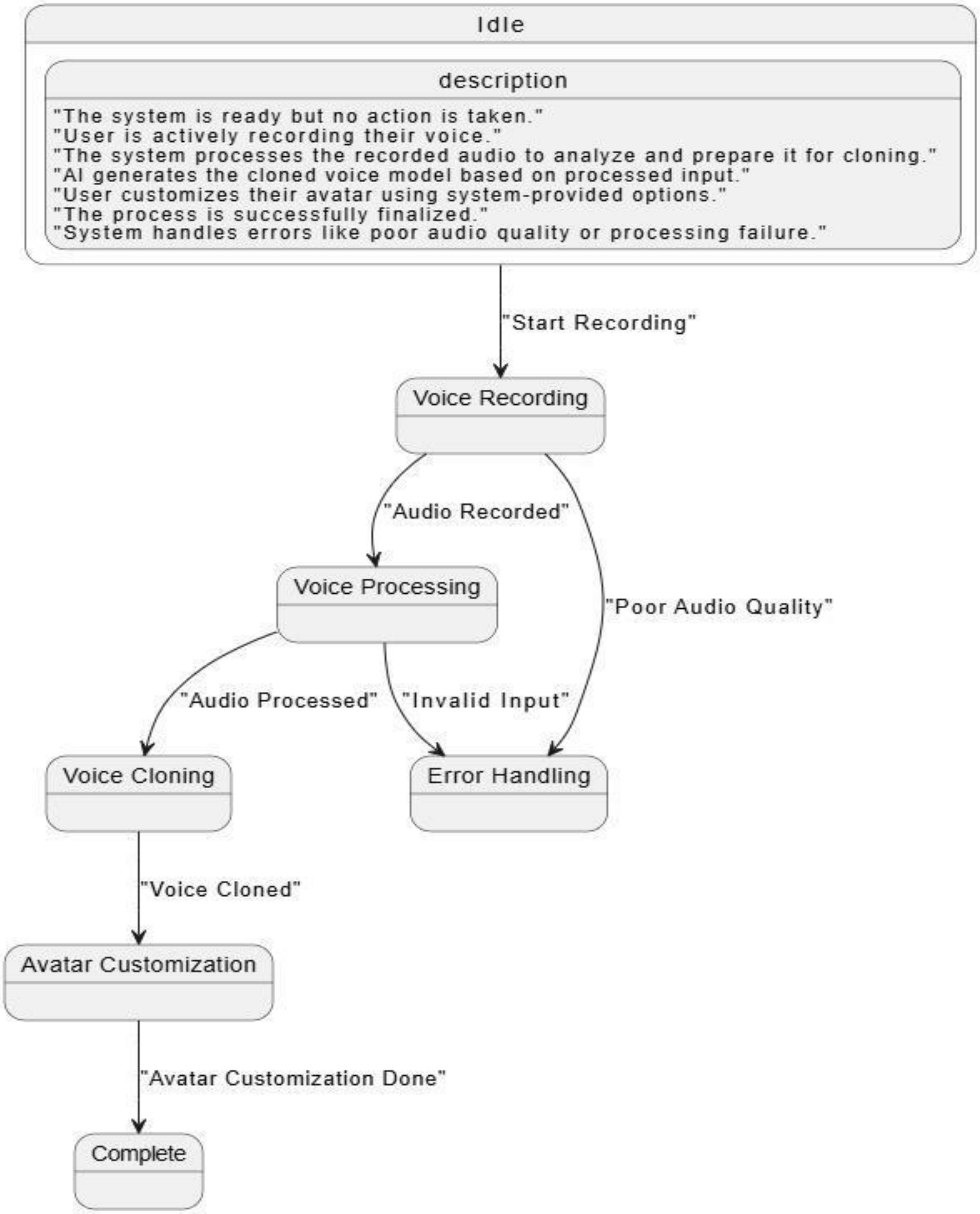


Figure 58: State Transition Diagram

4.9 Component Diagram

The component diagram outlines the physical artifacts of the system, showing how software components interact.

- **Frontend**: Flutter-based UI.
- **Backend**: Firebase for data management and cloud functions.
- **AI Modules**: TensorFlow for voice cloning, Unity for avatar rendering.

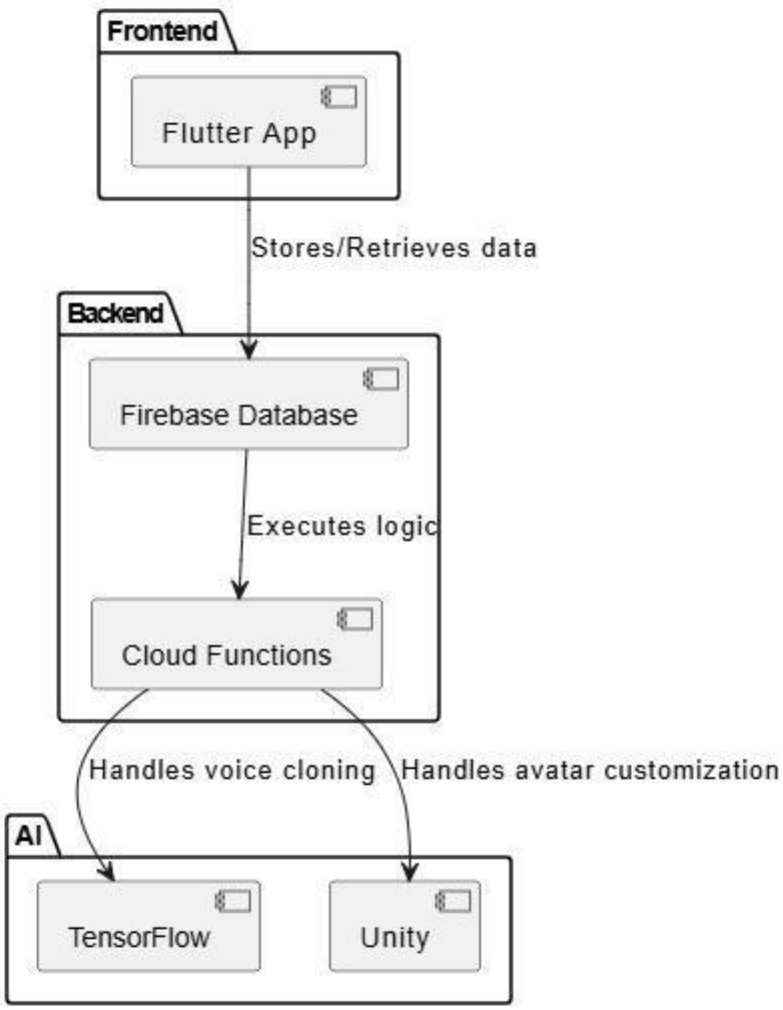


Figure 59: Component Diagram

4.10 Deployment Diagram

The deployment diagram represents the system's hardware and software distribution.

- **Mobile Devices**: Hosts the Flutter app for user interaction.
- **Cloud Infrastructure**: Includes Firebase for data storage and TensorFlow/Unity servers for processing.

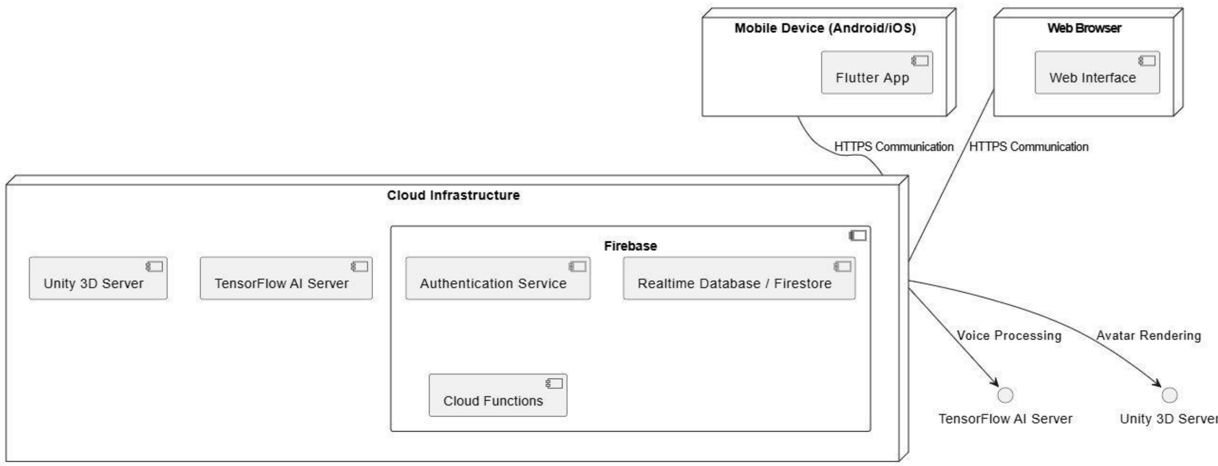


Figure 60: Deployment Diagram

CHAPTER 5: TESTING

5.1: Test case Specification

5.1.1: Test case Specification User Authentication (Sign-Up)

Positive test case:

Positive Test Case	
ID	TC_SIGNUP_001
Priority	High
Description	Verify user can successfully register a new account.
Reference	FR_01
Users	End-User
Pre-requisites	System must be online
Steps	1. Open sign-up page. 2. Enter valid name, email, and password. 3. Submit form.
Input	Valid username, email, and password.
Expected Result	User account is successfully created, and confirmation message is displayed.
Status	Tested, passed

Negative Test Case

Table 2: Sign Up negative Test Case

Negative Test Case	
ID	TC_SIGNUP_002
Priority	High
Description	Verify system prevents duplicate or invalid sign-up/login attempts.
Reference:	FR_01
Users	End user

Pre-requisites	System must be online
Steps	1. Open sign-up/login page. 2. Enter existing email, wrong or weak password. 3. Submit form.
Input	Duplicate email, invalid or weak password.
Expected Result	System displays appropriate error messages and prevents action.
Status	Tested, passed

5.1.2: Test case Specification User Authentication (Login)

Positive Test Case

Table 3: Login Positive Test Case

Positive Test Case	
ID	TC_ LOGIN_ 003
Priority	High
Description	Verify user can log in successfully with correct credentials.
Reference	FR_01
Users	End-User
Pre-requisites	System must be online and user account must exist.
Steps	1. Open login page. 2. Enter registered email and correct password. 3. Click Login.
Input	Registered email and correct password.
Expected Result	User is redirected to the dashboard or home page.
Status	Tested, passed

Negative Test Case

Table 4: Login Negative Test Case

Negative Test Case	
ID	TC_ LOGIN_ 004
Priority	High
Description	Verify system prevents login with incorrect credentials.
Reference:	FR_01
Users	End user
Pre-requisites	System must be online
Steps	1. Open login page. 2. Enter wrong email or incorrect password. 3. Click Login.
Input	Invalid or unregistered email and/or wrong password.
Expected Result	System displays error message: "Invalid credentials."
Status	Tested, passed

5.1.3: Test case Specification for Voice Cloning

Positive Test Case

Table 5: Voice Cloning Positive Test Case

Positive Test Case	
ID	TC_VOICECLONE_ 005
Priority	High
Description	Verify system successfully clones the user's voice from a clear recording.
Reference	FR_02
Users	End-User
Pre-requisites	System must be online and microphone must be functional.

Steps	1. Open voice cloning module. 2. Record a clear voice sample. 3. Submit the recording.
Input	Clear and valid voice sample.
Expected Result	System successfully processes and saves the cloned voice.
Status	Tested, passed

Negative Test Case

Table 6: Voice Cloning Negative Test Case

Negative Test Case	
ID	TC_ VOICECLONE_ 006
Priority	High
Description	Verify system rejects poor quality or incomplete voice samples.
Reference:	FR_02
Users	End user
Pre-requisites	System must be online and microphone must be functional.
Steps	1. Open voice cloning module. 2. Record noisy or very short audio. 3. Submit the recording.
Input	Noisy, incomplete, or silent audio sample.
Expected Result	System displays error message or prompts user to re-record.
Status	Tested, passed

AI Personalized Chatbot – Positive Test Case

5.1.4: Test case Specification for Real Time Interaction

Positive Test Case

Table 7: Real time Interaction Positive Test Case

Positive Test Case	
ID	TC_REALTIME_007
Priority	High
Description	Verify the system responds instantly to user's real-time text or voice input.
Reference	FR_03
Users	End-User
Pre-requisites	System must be active with input modules working.
Steps	1. Open the interface. 2. Provide real-time voice or text input. 3. Wait for response.
Input	Valid voice or text message.
Expected Result	The system responds instantly with an accurate reply.
Status	Tested, passed

Negative Test Case

Table 8: Real Time Interaction Negative Test Case

Negative Test Case	
ID	TC_REALTIME_008
Priority	High
Description	Verify the system handles unclear, unsupported, or empty inputs gracefully.
Reference	FR_03
Users	End-User
Pre-requisites	System must be active.

Steps	1. Open the interface. 2. Provide silence, gibberish, or unsupported input. 3. Observe response.
Input	Invalid voice or meaningless input.
Expected Result	System prompts for clarification or displays a suitable error message.
Status	Tested, passed

5.1.5: Test case Specification for Text to Speech

Positive Test Case

Table 9: Text to Speech Positive Test Case

Positive Test Case	
ID	TC_TextToSpeech_009
Priority	Medium
Description	Verify system converts entered text into audio using the cloned voice.
Reference	FR_04
Users	End-User
Pre-requisites	Cloned voice must already be generated.
Steps	1. Open TTS module. 2. Enter sample text. 3. Play the generated audio.
Input	Valid sentence or paragraph in English.
Expected Result	The system speaks the text using the user's cloned voice.
Status	Tested, passed

Negative Test Case

Table 10: Text To speech negative Test Case

Negative Test Case	
ID	TC_ TextToSpeech _010

Priority	Medium
Description	Verify system handles empty or excessively long text inputs.
Reference	FR_04
Users	End-User
Pre-requisites	Cloned voice must be available.
Steps	1. Open TTS module. 2. Leave the text field empty or enter very long input. 3. Submit.
Input	Empty text field or overly lengthy input.
Expected Result	System prompts user with a warning or error message.
Status	Tested, passed

5.1.6: Test case Specification for Accessibility for Speech- Impaired Users

Positive Test Case

Table 11: Accessibility for Speech- Impaired Users Positive Test Case

Positive Test Case	
ID	TC_ Access _011
Priority	High
Description	Verify system converts typed text into speech using the user's cloned voice.
Reference	FR_06
Users	Speech-impaired user
Pre-requisites	Text input module and cloned voice must be available.
Steps	1. Open accessibility interface. 2. Enter valid text. 3. Play generated audio.
Input	Clear English sentence (e.g., "Hello, how are you?").
Expected Result	Text is spoken using the cloned voice clearly.
Status	Tested, passed

Negative Test Case

Table 12: Accessibility for Speech- Impaired Users Negative Test Case

Negative Test Case	
ID	TC_ Access _012
Priority	High
Description	Verify system handles empty or unsupported inputs during accessibility use.
Reference	FR_06
Users	Speech-impaired user
Pre-requisites	Text-to-speech module must be loaded.
Steps	1. Open accessibility interface. 2. Leave text field empty or input gibberish.

	3. Click play.
Input	Empty or invalid text string.
Expected Result	System shows an error or requests the user to enter valid text.
Status	Tested, passed

5.1.7: Test case Specification for English Language

Positive Test Case

Table 13: English Language Positive Test Case

Positive Test Case	
ID	TC_ Language _013
Priority	Medium
Description	Validate system only English language input and provides accurate responses.
Reference	FR_07
Users	End user
Pre-requisites	System must be running.
Steps	1. Enter text input in English. 2. Submit input. 3. Observe system response.
Input	English sentence (e.g., “Tell me about the project.”).
Expected Result	System accepts and responds accurately to English input.
Status	Tested, passed

Negative Test Case

Table 14: English Language Negative Test Case

Negative Test Case	
ID	TC_ Language _014
Priority	Medium
Description	Verify the system rejects non-English language input with a suitable error message.
Reference	FR_07
Users	End user

Pre-requisites	System must be running.
Steps	1. Enter non-English text. 2. Submit input. 3. Observe system behavior.
Input	Non-English text (e.g., Urdu, Spanish, Arabic)
Expected Result	System rejects input and shows message like "Only English is supported."
Status	Tested, passed

5.1.8: Test case Specification for Virtual Meeting Participation

Positive Test Case

Table 15: Virtual Meeting Positive Test Case

Positive Test Case	
ID	TC_MEETING_015
Priority	High
Description	Verify the user's cloned voice is transmitted correctly in a virtual meeting.
Reference	FR_08
Users	End user
Pre-requisites	Cloned voice must be ready; stable internet and mic access must be granted.
Steps	1. Join the virtual meeting. 2. Speak using microphone. 3. Confirm voice is heard.
Input	Real-time speech through mic.
Expected Result	Other participants hear the user’s cloned voice clearly.
Status	Tested, passed

Negative Test Case

Table 16: Virtual Meeting Negative Test Case

Negative Test Case	
ID	TC_MEETING_016
Priority	High

Description	Verify system handles connection/microphone issues in meetings.
Reference	FR_08
Users	End user
Pre-requisites	Meeting module must be accessible.
Steps	1. Attempt to join with mic access denied or unstable internet. 2. Try to speak.
Input	User with no mic access or poor network.
Expected Result	System notifies the user of technical issue or prevents voice transmission.
Status	Tested, passed

5.1.9: Test case Specification for Avatar Customization

Positive Test Case

Table 17: Avatar Customization Positive Test Case

Positive Test Case	
ID	TC_AVATAR_017
Priority	Medium
Description	Verify that the user can select and apply an avatar from a list of available options.
Reference	FR_09
Users	End user
Pre-requisites	System must load all available avatar options successfully.
Steps	1. Open avatar selection screen. 2. Choose one avatar from the displayed list. 3. Confirm the selection.
Input	Selected avatar from available options.
Expected Result	The selected avatar is applied and shown during interaction.

Status	Tested, passed

Negative Test Case
Table 18: Avatar Customization Negative Test Case

Negative Test Case	
ID	TC_AVATAR_018
Priority	Medium
Description	Verify the system handles failed avatar selection or missing selection.
Reference	FR_09
Users	End user
Pre-requisites	Avatar selection interface must be available.
Steps	1. Open avatar selection screen. 2. Attempt to proceed without selecting any avatar. 3. Try to continue.
Input	No avatar selected.
Expected Result	System prompts user to select an avatar before proceeding.
Status	Tested, passed

5.1.10: Test case Specification for User Feedback Mechanism

Positive Test Case
Table 19: User Feedback Mechanism Positive Test Case

Positive Test Case	
ID	TC_FEEDBACK _019
Priority	Low
Description	Verify that the user can submit feedback successfully through the feedback form.
Reference	FR_10
Users	End user
Pre-requisites	Feedback form must be accessible and connected to the database.
Steps	1. Navigate to the feedback section. 2. Enter valid comments in the form. 3. Submit.
Input	Properly filled feedback text field.
Expected Result	System saves the feedback and displays a success

	message.
Status	Tested, passed

Negative Test Case

Table 20: User Feedback Positive Test Case

Negative Test Case	
ID	TC_FEEDBACK _020
Priority	Low
Description	Verify that the system does not accept empty feedback submissions.
Reference	FR_10
Users	End user
Pre-requisites	Feedback module must be live.
Steps	1. Navigate to the feedback form. 2. Leave the comment field blank. 3. Try to submit.
Input	Empty comment field.
Expected Result	System displays a validation error and does not submit feedback.
Status	Tested, passed

Black Box Testing

5.2 Black Box Testing

5.2.1 Equivalence Partitioning

5.2.1.1 Equivalence Partitioning for Sign-up

Table 21: Equivalence Partitioning for Sign-up

Variables	Valid Class	Invalid Class
Username	3–20 characters, includes alphabets (a–z, A–Z), digits (0–9), allowed symbols	<3 or >20 characters, special characters not allowed, empty field
Email	Valid email format (e.g., name@domain.com)	Missing “@” or “.com”, contains spaces, empty field
Password	6–20 characters, includes letters, numbers, and symbols	<6 or >20 characters, no symbols, empty field

Confirm Password	Matches the original password exactly	Does not match password, empty field

5.2.1.2 Equivalence Partitioning for Login

Table 22: Equivalence Partitioning for Login

Variables	Valid Class	Invalid Class
Email	Registered email with correct format	Unregistered email, wrong format, empty field
Password	Correct password for registered email	Wrong password, empty password field

5.2.1.3 Equivalence Partitioning for Voice Cloning

Table 23: Equivalence Partitioning for Voice Cloning:

Variables	Valid Class	Invalid Class
Audio Sample	Clear voice ≥10 seconds, in selected language (English/Urdu)	<5 seconds, noisy, wrong language, silence, empty input
File Format	.wav, .mp3, .m4a	.txt, .pdf, corrupted file, unsupported format
Language Match	Input language matches selected cloning language	Language mismatch (e.g., Urdu voice when English selected)

5.2.1.4 Equivalence Partitioning for Real-Time Interaction

Table 24: Equivalence Partitioning for Real-Time Interaction

Variables	Valid Class	Invalid Class
User Input	Recognizable voice or clear text input	Silence, noise, gibberish, empty string
Connection	Stable internet connection	Disconnected, laggy, low bandwidth

5.2.1.5 Equivalence Partitioning for Text-to-Speech

Table 25: Equivalence Partitioning for Text-to-Speech

Variables	Valid Classes	Invalid Classes
Text Input	Plain, short sentences (≤500 characters)	>500 characters, emojis only, symbols only

Voice Model	Pre-generated and active cloned voice	Missing or uninitialized voice model

5.2.1.6 Equivalence Partitioning for Accessibility for Speech-Impaired Users

Table 26: Equivalence Partitioning for Accessibility Feature

Variables	Valid Classes	Invalid Classes
Input Type	Text input by user	Unsupported inputs (e.g., image, binary, blank)
Response Type	Cloned voice playback	No response, system failure, wrong voice

5.2.1.7 Equivalence Partitioning for English Language Support

Table 27: Equivalence Partitioning for English Language Support

Variables	Valid Classes	Invalid Classes
Language Input	English voice or text input	Urdu or other language input, mixed-language input
Pronunciation	Clear, fluent English pronunciation	Slurred, mumbled, or heavily accented speech
Output Language	System response in clear English (cloned voice)	Non-English output, silence, or incorrect voice

5.2.1.8 Equivalence Partitioning for Virtual Meeting Participation

Table 28: Equivalence Partitioning for Virtual Meeting Participation

Variables	Valid Classes	Invalid Classes
Audio Input	Clear, real-time voice in native language	Distorted audio, dropped packets, wrong language
Connectivity	Stable internet connection	Disconnected, jittery, no mic access
Voice Output	Participants hear cloned voice clearly	No voice playback, robotic distortion, wrong voice model

5.2.1.9 Equivalence Partitioning for Avatar Customization

Table 29: Equivalence Partitioning for Avatar Customization

Variables	Valid Classes	Invalid Classes

Avatar Inputs	Acceptable avatar presets and custom settings	Missing fields, invalid values, unsupported file types
Image Upload (if any)	Valid JPG/PNG files ≤5MB	Corrupted file, unsupported format, size >5MB

5.2.1.10 Equivalence Partitioning for User Feedback Mechanism

Table 30: Equivalence Partitioning for User Feedback Mechanism

Variables	Valid Classes	Invalid Classes
Feedback Content	10–500 characters with relevant text	Empty feedback, spam content, repeated characters only
Rating Value	Integer value from 1 to 5	0, negative numbers, symbols, non-integer inputs

Boundary Value Analysis

5.2.2 Boundary Value Analysis

5.2.2.1 Boundary Value Analysis for Sign-up

Table 31: Boundary Value Analysis for Sign-Up

Test Case	Invalid (min-1)	Valid (min)	Valid (+min)	Valid (-max)	Valid (max)	Invalid (max+1)
Username Length	2 chars	3 chars	4 chars	19 chars	20 chars	21 chars
Password Length	5 chars	6 chars	7 chars	19 chars	20 chars	21 chars
Email Address	Invalid format	Valid format	–	–	Valid format	Invalid format

5.2.2.2 Boundary Value Analysis for Login

Table 32: Boundary Value Analysis for Login

Test Case	Invalid (min-1)	Valid (min)	Valid (+min)	Valid (-max)	Valid (max)	Invalid (max+1)
Email Address	Invalid format	Valid format	–	–	Valid format	Invalid format
Password	5 chars	6 chars	7 chars	19 chars	20 chars	21 chars

5.2.2.3 Boundary Value Analysis for Voice Cloning

Table 33: Boundary Value Analysis for Voice Cloning

Test Case	Invalid (min-1)	Valid (min)	Valid (+min)	Valid (-max)	Valid (max)	Invalid (max+1)
Audio Duration	4 sec	5 sec	10 sec	29 sec	30 sec	31 sec
File Size (MB)	–	0.5MB	1MB	19 chars	49MB	5MB

5.2.2.4 Boundary Value Analysis for Real-Time Interaction

Table 34: Boundary Value Analysis for Real-Time Interaction

Test Case	Invalid (min-1)	Valid (min)	Valid (+min)	Valid (-max)	Valid (max)	Invalid (max+1)
Input Words	0 words	1 word	2 words	199 words	200 words	201 words
Response Time	–	0.1 sec	1 sec	2.4 sec	2.5 sec	2.6 sec

5.2.2.5 Boundary Value Analysis for Text-to-Speech

Table 35: Boundary Value Analysis for Text-to-Speech

Test Case	Invalid (min-1)	Valid (min)	Valid (+min)	Valid (-max)	Valid (max)	Invalid (max+1)
Text Length	0 characters	1 character	2 characters	499 characters	500 characters	501 characters
Playback time	–	1 sec	2 sec	29 sec	30 sec	31 sec (cut-off)

5.2.2.6 Boundary Value Analysis for Accessibility for Speech-Impaired Users

Table 36: Boundary Value Analysis for Accessibility Input

Test Case	Invalid (min-1)	Valid (min)	Valid (+min)	Valid (-max)	Valid (max)	Invalid (max+1)
Text Length	0 characters	1 character	2 characters	499 characters	500 characters	501 characters
Response Delay	–	1 sec	2 sec	2.9 sec	3 sec	3.1 sec

5.2.2.7 Boundary Value Analysis for English Language

Table 37: Boundary Value Analysis for English Language

Test Case	Inv (min-1)	Val (min)	Val (+min)	Val (-max)	Val (max)	Inv(max+1)
Sentence Length	0 words	1 word	2 words	29 words	30 words	31+ words
Language Detected	Non-English	Basic English	Fluent English	–	Clear English	Mixed/foreign
Pronunciation Clarity	Unclear	Understandable	Neutral accent	–	Clear accent	Garbled speech

5.2.2.8 Boundary Value Analysis for Virtual Meeting Participation

Table 38: Boundary Value Analysis for Virtual Meeting Participation

Test Case	Inv (min-1)	Val (min)	Val (+min)	Val (-max)	Val (max)	Inv (max+1)
Meeting Duration	–	1 min	2 min	59 min	60 min	61+ min
Audio Latency (ms)	301 ms	300 ms	250 ms	100 ms	50 ms	–

5.2.2.9 Boundary Value Analysis for Avatar Customization

Table 39: Boundary Value Analysis for Avatar Customization

Test Case	Inv (min-1)	Val (min)	Val (+min)	Val (-max)	Val (max)	Inv (max+1)
Custom Fields Selected	2 options	3 options	4 options	9 options	10 options	11+ options
Image Upload Size (MB)	>5 MB	5 MB	4 MB	2 MB	1 MB	>5 MB

5.2.2.10 Boundary Value Analysis for User Feedback Mechanism

Table 40: Boundary Value Analysis for User Feedback Mechanism

Test Case	Inv (min-1)	Val (min)	Val (+min)	Val (-max)	Val (max)	Inv (max+1)
Feedback Length	9 chars	10 chars	11 chars	499 chars	500 chars	501+ chars
Rating (1–5 Scale)	0	1	2	4	5	6 or more

Decision Table

5.2.3 Decision Table

5.2.3.1 Decision Table for Sign-Up

Table 41: Decision Table for Sign-Up

Condition	Test1	Test2	Test3	Test4
Username Availability (T/F)	T	T	F	F
Email Format Valid (T/F)	T	F	T	F
Password Length Valid (T/F)	T	F	T	F
Action	TRUE	FALSE	FALSE	FALSE

5.2.3.2 Decision Table for Login

Table 42: Decision Table for Login

Condition	Test1	Test2	Test3	Test4
Username Exists (T/F)	T	T	F	F
Password Correct (T/F)	T	F	T	F
Account Active (T/F)	T	T	T	F
Action	TRUE	FALSE	FALSE	FALSE

5.2.3.3 Decision Table for Voice Cloning

Table 43: Decision Table for Voice Cloning

Condition	Test1	Test2	Test3	Test4
Voice Data Available (T/F)	T	T	F	F
Microphone Functional (T/F)	T	F	T	F
Audio Quality Acceptable (T/F)	T	T	T	F
Action	TRUE	FALSE	FALSE	FALSE

5.2.3.4 Decision Table for Real-Time Interaction

Table 44: Decision Table for Real-Time Interaction

Condition	Test1	Test2	Test3	Test4
User Authenticated (T/F)	T	T	F	T

System Online (T/F)	T	F	T	T
Input Provided (T/F)	T	T	T	F
Action	TRUE	FALSE	FALSE	FALSE

5.2.3.5 Decision Table for Text-to-Speech

Table 45: Decision Table for Text-to-Speech

Condition	**Test1**	**Test2**	**Test3**	**Test4**
Text Input Available (T/F)	T	T	F	F
Cloned Voice Available (T/F)	T	F	T	F
TTS Engine Functional (T/F)	T	T	T	F
Action	TRUE	FALSE	FALSE	FALSE

5.2.3.6 Decision Table for Accessibility for Speech-Impaired Users

Table 46: Decision Table for Accessibility for Speech-Impaired Users

Condition	**Test1**	**Test2**	**Test3**	**Test4**
Text Input Enabled (T/F)	T	T	F	T
Cloned Voice Available (T/F)	T	F	T	F
Voice Output Functional (T/F)	T	T	T	F
Action	TRUE	FALSE	FALSE	FALSE

5.2.3.7 Decision Table for English Language Support

Table 47: Decision Table for English Language Support

Condition	**Test1**	**Test2**	**Test3**	**Test4**
Language Selected is English (T/F)	T	T	F	F
Language Engine Available (T/F)	T	F	T	F
Cloned Voice Available (T/F)	T	T	T	F
Action	TRUE	FALSE	FALSE	FALSE

5.2.3.8 Decision Table for Virtual Meeting Participation

Table 48: Decision Table for Virtual Meeting Participation

Condition	**Test1**	**Test2**	**Test3**	**Test4**
Meeting Module Active (T/F)	T	T	F	T

Internet Stable (T/F)	T	F	T	T
Cloned Voice Available (T/F)	T	T	T	F
Action	TRUE	FALSE	FALSE	FALSE

5.2.3.9 Decision Table for Avatar Customization (Prebuilt)

Table 49: Decision Table for Avatar Customization

Condition	Test1	Test2	Test3	Test4
Prebuilt Avatars Available (T/F)	T	T	F	F
Avatar Module Active (T/F)	T	F	T	F
User Selection Made (T/F)	T	T	T	F
Action	TRUE	FALSE	FALSE	FALSE

5.2.3.10 Decision Table for User Feedback Mechanism

Table 50: Decision Table for User Feedback Mechanism

Condition	Test1	Test2	Test3	Test4
Feedback Form Accessed (T/F)	T	T	F	F
User Authenticated (T/F)	T	F	T	F
Feedback Submitted (T/F)	T	T	F	F
Action	TRUE	FALSE	FALSE	FALSE

5.2.4 State transition Testing:

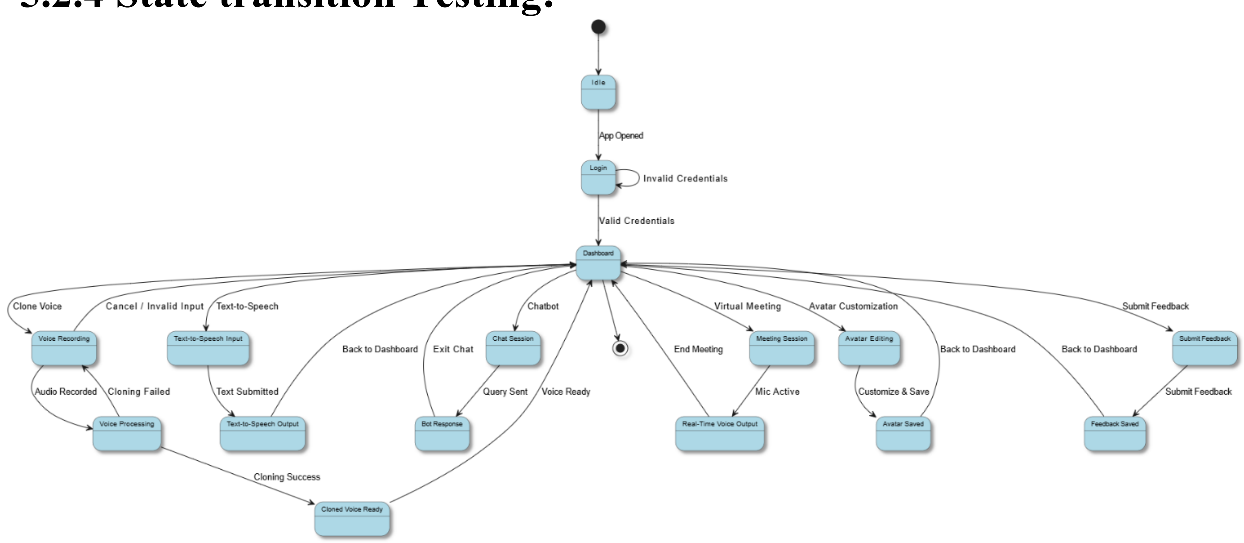


Figure 61: State Transition Testing

Table 51: State Transition Diagram

Current State	Sign Up	Login	Dashboard	Voice Cloning	Real-Time interaction	TTS	Avatar customize	User Feedback
S1: Sign Up	-	-	-	-	-	-	-	-
S2: Login	S1	-	-	-	-	-	-	-
S3: Dashboard	-	S2	-	-	-	-	-	-
S4: Voice cloning	-	-	S3	-	-	-	-	-
S5: Real-Time Int	-	-	-	S4	-	-	-	-
S6: TTS	-	-	-	-	S5	-	-	-
S7: Avatar customize	-	-	-	-	-	S6	-	-
S8: User Feedback	-	-	-	-	-	-	S7	-

Use-Case Testing

5.2.5.1 Use Case Testing for Sign-up

Table 52: Use Case Testing for Sign-Up

Field	Details
Description	User shall open system for signup
Test Case ID	TC_SIGNUP_001
Actors	Guest, System
Main Scenario	
Serial No.	Steps
1	Open the System
2	Navigate to the signup page
3	Enter valid user details
4	Submit the form
Extensions	2a. Password didn't match 3a. Invalid email format 4a. Username already exists

5.2.5.2 Use Case Testing for Login

Table 53: Use Case Testing for Login

Field	Details
Description	User shall log in to the system using credentials
Test Case ID	TC_LOGIN_002
Actors	User, System
Main Scenario	
Serial No.	**Steps**
1	Open the system
2	Navigate to the login page
3	Enter username and password
4	Click the login button
Extensions	3a. Incorrect username or password 4a. Account is not active

5.2.5.3 Use Case Testing for Voice Cloning

Table 54: Use Case Testing for Voice Cloning

Field	Details
Description	User shall record and clone their voice
Test Case ID	TC_VOICECLONE_003
Actors	User, System
Main Scenario	
Serial No.	**Steps**
1	Navigate to the voice cloning module
2	Record voice sample
3	Submit the sample for cloning
4	System generates and plays back cloned voice
Extensions	2a. Microphone access denied 3a. Poor audio quality 4a. Cloning failed

5.2.5.4 Use Case Testing for Real-Time Interaction

Table 55: Use Case Testing for Real-Time Interaction

Field	**Details**
Description	System shall respond to user input in real time
Test Case ID	TC_REALTIME_004
Actors	User, System
Main Scenario	
Serial No.	**Steps**
1	Launch the application
2	Provide voice or text input
3	System processes the input
4	System returns a real-time response
Extensions	2a. Input not recognized 3a. System delay or offline

5.2.5.5 Use Case Testing for Text-to-Speech

Table 56: Use Case Testing for Text-to-Speech

Field	**Details**
Description	System shall convert user text input into speech using cloned voice
Test Case ID	TC_TTS_005
Actors	User, System
Main Scenario	
Serial No.	Steps
1	Access text-to-speech feature
2	Enter desired text input
3	Click the convert/play button
4	System generates and plays speech using cloned voice
Extensions	2a. Text field is empty 4a. Cloned voice not available

5.2.5.6 Use Case Testing for Accessibility for Speech-Impaired Users

Table 57: Use Case Testing for Accessibility for Speech-Impaired Users

Field	Details
Description	Speech-impaired user shall interact using text input and receive voice output
Test Case ID	TC_ACCESS_006
Actors	User (Speech-Impaired), System
Main Scenario	
Serial No.	**Steps**
1	Open the system and enable accessibility mode
2	Enter message in text input field
3	Submit the message
4	System responds with audio in cloned voice
Extensions	2a. Keyboard not responding 4a. Cloned voice not available

5.2.5.7 Use Case Testing for English Language

Table 58: Use Case Testing for English Language

Field	Details
Description	User shall communicate in English and receive English responses
Test Case ID	TC_ENGLISH_007
Actors	User, System
Main Scenario	
Serial No.	**Steps**
1	Launch the system
2	Select English as preferred language
3	Provide voice or text input in English
4	System responds in English
Extensions	2a. Language selection failed 3a. Input not recognized as English

5.2.5.8 Use Case Testing for Virtual Meeting Participation

Table 59: Use Case Testing for Virtual Meeting Participation

Field	Details
Description	User shall join and communicate in a virtual meeting using cloned voice
Test Case ID	TC_MEETING_008
Actors	User, System, Other Participants
Main Scenario	
Serial No.	**Steps**
1	Open and launch the system
2	Join a scheduled virtual meeting
3	Speak or input message
4	System transmits the message using the cloned voice
Extensions	2a. Network issue during meeting 4a. Cloned voice not functioning

5.2.5.9 Use Case Testing for Avatar Customization

Table 60: Use Case Testing for Avatar Customization

Field	Details
Description	User shall select a prebuilt avatar from available options
Test Case ID	TC_AVATAR_009
Actors	User, System
Main Scenario	
Serial No.	**Steps**
1	Open the avatar customization module
2	View the list of prebuilt avatars
3	Select an avatar
4	System saves and applies the selected avatar
Extensions	2a. Avatar list failed to load 3a. Avatar selection failed

5.2.5.10 Use Case Testing for User Feedback Mechanism

Table 61: Use Case Testing for User Feedback Mechanism

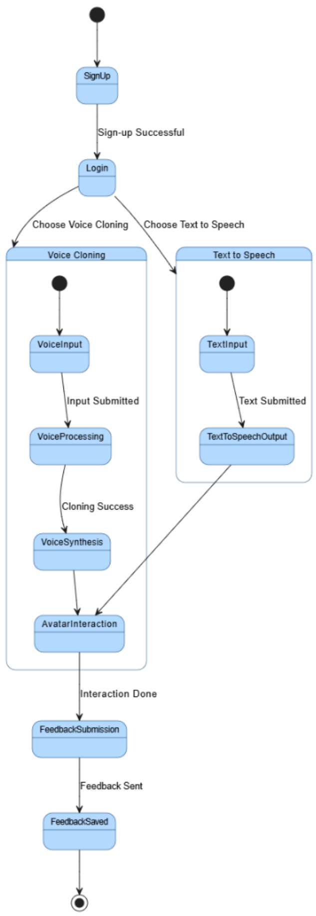
Field	Details
Description	User shall submit feedback through the system
Test Case ID	TC_FEEDBACK_010
Actors	User, System
Main Scenario	
Serial No.	**Steps**
1	Navigate to the feedback section
2	Enter feedback into the form
3	Submit the form
4	System confirms successful submission
Extensions	2a. Feedback field is empty 3a. Submission failed due to network error

White Box Testing

5.3 White Box Testing

5.3.1 Cyclomatic Complexity

Cyclomatic Complexity is a software metric used to measure the logical complexity of a program. It helps determine the number of independent paths and is essential for ensuring adequate test coverage.



5.3.1.1 Cyclomatic Complexity for State Transition Diagram:

5.3.1.2 Cyclomatic Complexity

Table 62: Parameters for Cyclomatic Complexity

Parameter	Value	Description
N	12	Number of nodes (states)
E	16	Number of edges (transitions)
P	1	Number of connected components

5.3.1.3 Cyclomatic Complexity Formula

Formula:

$$M = E - N + 2P$$

Path Coverage

- Path 1: Start → Sign-up (Yes) → Login (Yes) → Dashboard → Enter Input → Process Input → Synthesize Voice → Avatar Animation → Avatar Interaction
- Path 2: Start → Sign-up (No)

- Path 3: Start → Sign-up (Yes) → Login (No)

Statement Coverage
The Statement Coverage is 100% because all the nodes (states) are covered in one go.

Branch Coverage
The Branch Coverage is 100% because all decision outcomes (Yes/No branches) are tested.

CHAPTER 6: TOOLS AND TECHNIQUES

Tools and Techniques

6.1 Languages Used in Development

The development of the **Voca-Twin** application involved the use of the following programming languages, each chosen to support specific aspects of the system effectively:

- **Dart**: Used for building the frontend using Flutter, Dart enabled cross-platform development for Android, iOS, and web. It provided a consistent and responsive user interface with smooth performance across all supported devices.

- **Python**: Utilized for implementing the artificial intelligence modules, including voice cloning and natural language processing. Python's compatibility with machine learning frameworks like TensorFlow made it ideal for training and deploying the voice synthesis and emotion-aware response models.

6.2 Applications and Tools

The development of the Voca-Twin system involved a diverse set of applications and tools, carefully selected to streamline the development process, support AI functionalities, and ensure a smooth user experience across platforms:

- **Flutter (in Visual Studio Code)**: Flutter was used to build the cross-platform frontend of the application, enabling deployment on Android, iOS, and web from a single codebase. Visual Studio Code served as the primary development environment due to its lightweight interface and powerful extension support.

- **Firebase**: Used as the backend platform, Firebase provided essential services such as real-time database, user authentication, and cloud storage. Its integration simplified backend operations while maintaining scalability and security.

- **Flask with Python**: Flask, a lightweight Python web framework, was used to create the server-side API layer for handling AI-related tasks. It facilitated interaction between the frontend and Python-based machine learning modules.

- **Figma**: Figma was employed for designing the user interface and user experience (UI/UX) components of the application. It allowed collaborative, cloud-based design work and rapid prototyping.

- **Real-Time Voice Cloning (by CorentinJ)**: The voice cloning functionality was implemented using the open-source project **Real-Time-Voice-Cloning.** This model was used for training and generating voice profiles that mimic the user's speech, tone, and emotional inflections.

6.3 Libraries and Extensions

Several libraries and extensions were integrated into the Voca-Twin system to support frontend development, machine learning tasks, and backend functionality. These tools enhanced performance, accelerated development, and enabled complex features like real-time voice cloning and dynamic UI behavior.

Frontend (Flutter / Dart)

- **Provider**: For state management across the application to ensure efficient UI updates.
- **Flutter Hooks**: Simplified widget lifecycle management and cleaner code structure.
- **Firebase Core & Auth**: For integrating Firebase authentication and backend services.
- **http**: Used for sending and receiving data between the app and backend API endpoints.
- **Flutter WebRTC** *(if applicable)*: For real-time communication features in virtual meetings.

Backend (Python / Flask)

- **Flask**: A micro web framework used to handle HTTP requests and run the backend services.
- **TensorFlow**: For implementing and running machine learning models used in voice synthesis and processing.
- **NumPy & SciPy**: For numerical operations and signal processing during voice training.
- **Librosa**: For analyzing and processing audio data, particularly in feature extraction from voice samples.
- **PyTorch** *(used indirectly via the voice cloning repo)*: The Real-Time Voice Cloning model utilizes PyTorch for deep learning tasks.
- **Soundfile / Wave**: To handle audio input/output operations during preprocessing and synthesis.

Browser & Development Extensions

- **Flutter & Dart Plugins** (VS Code): Enabled smart autocompletion, hot reload, and debugging tools for Flutter development.
- **Firebase Tools CLI**: Used for deploying and managing Firebase services directly from the terminal.

CHAPTER 7: SUMMARY AND CONCLUSION

Summary

Voca-Twin is an AI virtual assistant system designed to revolutionize user interaction through personalized and engaging experiences. Using coretinJ for voice cloning and sadtalker for avatar generation, the system creates a digital twin of the user. It focuses on an AI-based approach for intelligent, adaptive responses, aiming to redefine virtual assistants.

A key feature is voice cloning via coretinJ. This captures unique vocal traits to create a digital voice twin. The cloned voice is used for real-time communication, making the AI assistant sound like the user. This promotes a sense of connection and familiarity in interactions for applications like chatbots and virtual meetings.

Voca-Twin also includes avatar customization and generation with sadtalker. Users can create and personalize digital avatars representing their virtual companion. These avatars sync with the voice and show emotion, enhancing realism and allowing users to customize the AI assistant's look for a more engaging experience.

Real-time interaction is central to Voca-Twin. The system processes voice or text input and responds quickly, ensuring smooth conversations. This low-latency performance is vital for engaging dialogue during questions or virtual meetings, distinguishing Voca-Twin as a responsive AI companion.

The Voca-Twin system supports communication in the English language. Users interact via voice or text in English, receiving responses in the cloned voice. This ensures accurate communication within the specified language, catering to English-speaking users and optimizing the personalized experience.

Finally, Voca-Twin includes accessibility and utility features. It supports text input for speech-impaired users, providing responses in their cloned voice. Users can also use their cloned voice in virtual meetings. These features, along with potential uses in customer support and reminders, show Voca-Twin's potential for inclusive, personalized, and engaging AI interactions.

Conclusion

In conclusion, Voca-Twin represents a significant step forward in AI virtual assistant technology. The project successfully developed a system integrating personalized features like voice cloning and avatars to create more human-like interactions. By focusing on an AI-based approach, Voca-Twin moves beyond the limitations of existing virtual assistants, offering a more engaging and personalized user experience. The successful implementation of core functionalities demonstrates the project's technical achievements and potential impact on the field of AI interaction.

The core functionality of voice cloning, implemented using coretinJ, was a key achievement. This allows users to create a digital voice twin, enabling the AI assistant to communicate in a familiar voice. The successful replication of vocal characteristics for real-time interaction provides a level of personalization not commonly found in current systems. This capability is foundational to Voca-Twin's goal of creating a truly personalized virtual companion.

The integration of avatar generation using sadtalker further enhances the user experience. The ability to create and customize a visual representation of the AI assistant adds a crucial layer of engagement. These avatars, synchronized with the cloned voice, contribute to more realistic and emotionally resonant interactions. This visual component is essential in making the virtual assistant feel more like a companion.

Real-time interaction was a primary objective and a successful outcome of the Voca-Twin project. The system's ability to process input and respond instantly ensures smooth and dynamic conversations. This low-latency performance is critical for maintaining user engagement and provides a responsive interaction experience. The successful implementation of real-time processing highlights the system's technical robustness.

Voca-Twin was developed with a focus on providing a high-quality experience in the English language. The system effectively handles English voice and text input, generating responses in the user's cloned voice. This targeted language support ensures accuracy and a consistent experience for English-speaking users. The project successfully demonstrated the system's capabilities within this language constraint.

Finally, the project highlighted the potential of Voca-Twin for various applications, including accessibility and virtual communication. Features supporting text input for speech-impaired users and enabling participation in virtual meetings underscore the system's versatility. The successful development and integration of these features position Voca-Twin as a promising AI-powered virtual assistant with the capacity to deliver inclusive and highly personalized interactions.

CHAPTER 8: USER MANUAL

Introduction of App

Voca Twin is an easy-to-use app that lets you create your own AI voice. It copies your real voice and then speaks just like you. You can use it to read messages, answer questions, or talk in your own voice without even speaking. The app is designed to be simple and fun. Just follow the steps in this guide to get started.

Step By Step Guide

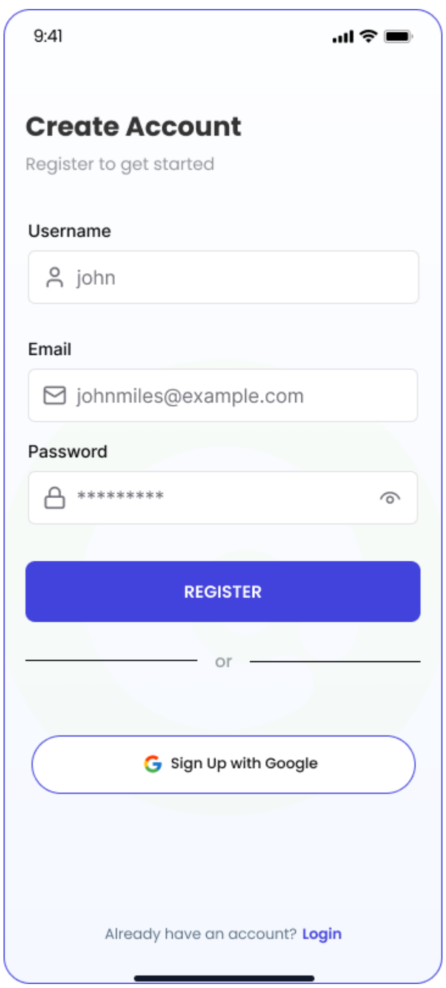
8.1 Sign Up

After opening the Voca Twin app, the first screen you'll see is the Sign-Up page. This is where you can create your new account.

Steps to Sign Up:

1. Enter your Username.
2. Type your Email address.
3. Create a Password.
4. Tap the Register button.

You can also choose to Sign Up with Google or Sign Up with Apple for faster access.



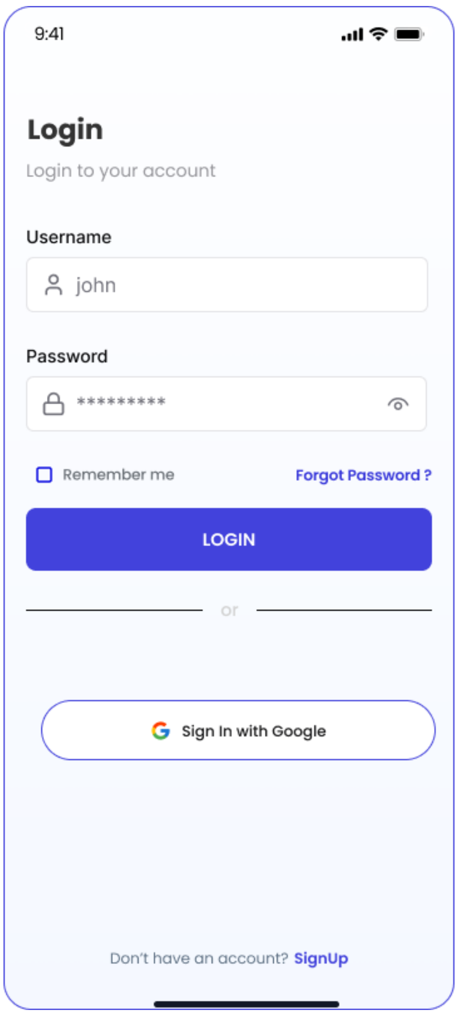
8.2 Login:

If you already have an account, you'll be directed to the Login Page instead of the Sign Up screen.

Steps to Log In:

1. Enter your email address.
2. Type your password.
3. Tap the Login button.

You can also log in using Google or Apple if you signed up that way.
Once logged in successfully, you'll be taken straight to the Welcome Page, where you can start your voice journey.
No need to sign up again just log in and jump right back in.



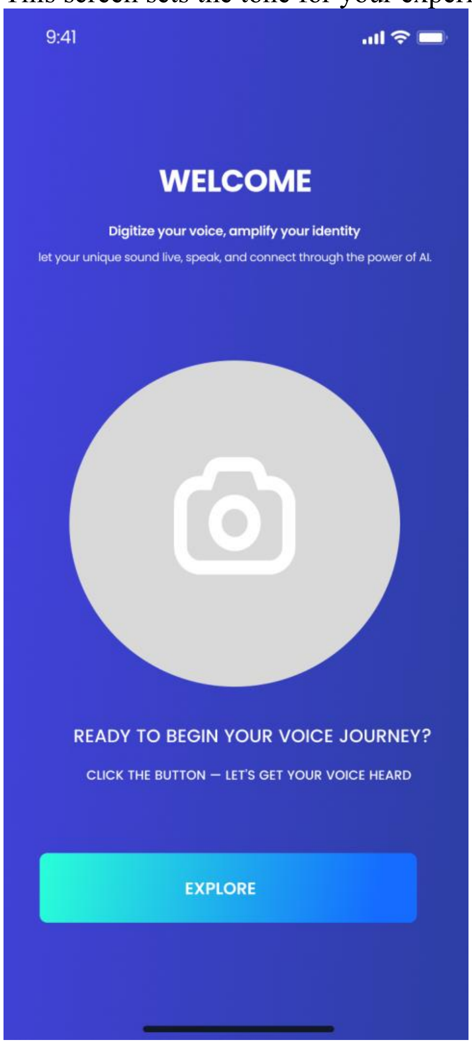
8.3 Welcome page:

After signing up or logging in, you'll land on the Welcome Page. This is where your Voca Twin journey officially begins.

Here's what you'll find on this screen:

- A warm welcome message.
- A line that encourages you to digitize your voice and express your identity.
- A bold microphone icon to represent your voice.
- A clear "Explore" button to move forward and begin your voice setup.

This screen sets the tone for your experience. Inviting you to explore how your voice can come to life using AI.



8.4 Home Page:

Once you log in, you'll land on the Home Page — your central hub to begin your voice journey.
What you can do on the Home Page:

1. Tap Record & Save Avatar to record your voice and create your own voice avatar.

This is the first step your avatar won't be active until you've saved a voice.

2. View Celebrity Avatar Images displayed below — these are just visuals to explore, with no built-in voices yet.

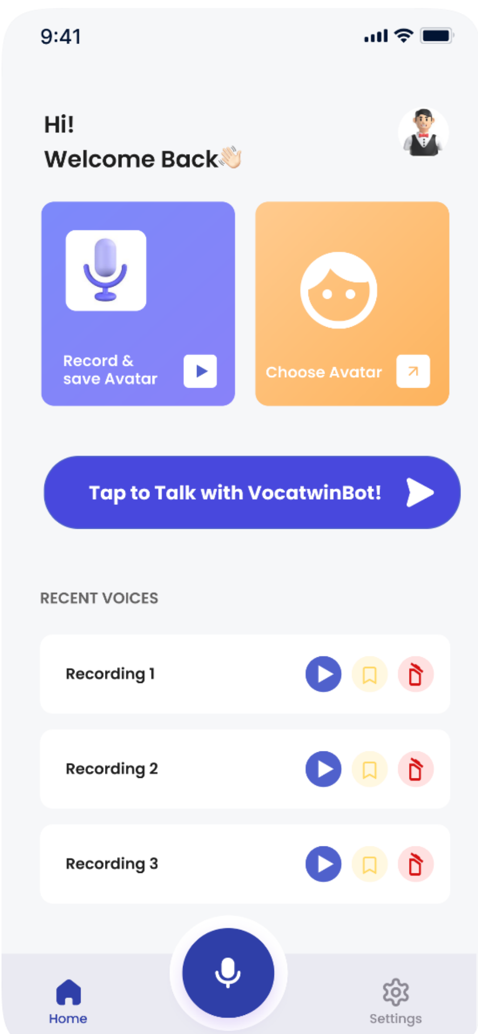
3. In the Recent Voices section, you'll see your saved recordings here once you start creating them.

For now, it's empty until you record and save your first voice.
At the bottom of the screen:

- Tap the Microphone icon to quickly start a new voice recording.

- Use the Settings icon to adjust your preferences.

This page starts off clean it's waiting for *your* voice to bring it to life.



8.5 VocaTwinBot Page:

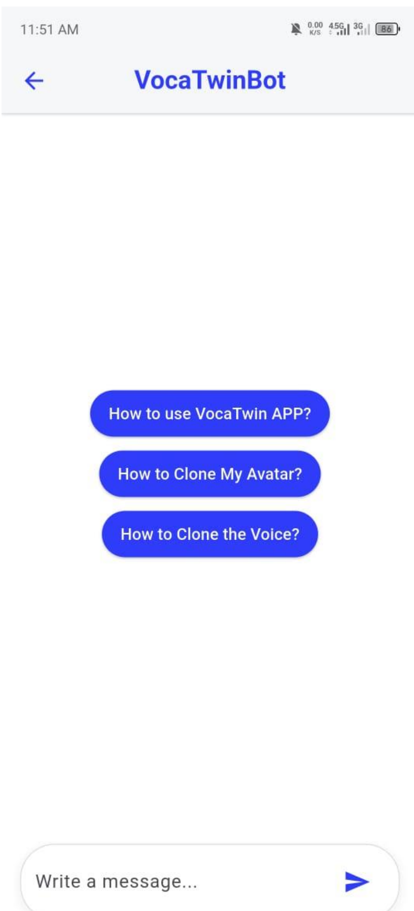
Once you open the bot, you'll land on the VocaTwinBot Page — your guide to using the app's voice an avatar cloning features.

What you can do on the VocaTwinBot Page:

1. **Tap "How to use VocaTwin APP?"**
   Learn the basic functions of the app, how to navigate, and where to begin your voice journey.
2. **Tap "How to Clone My Avatar?"**
   Get step-by-step instructions on creating your personalized avatar using your photo or video.
3. **Tap "How to Clone the Voice?"**
   Understand how to record, save, and apply your own voice to your avatar for a more personal experience.

At the bottom of the screen:

• Type a message to ask any question related to the app and get instant help.
• Tap the Send icon to interact directly with the bot.



8.6 Voice Recording Page:

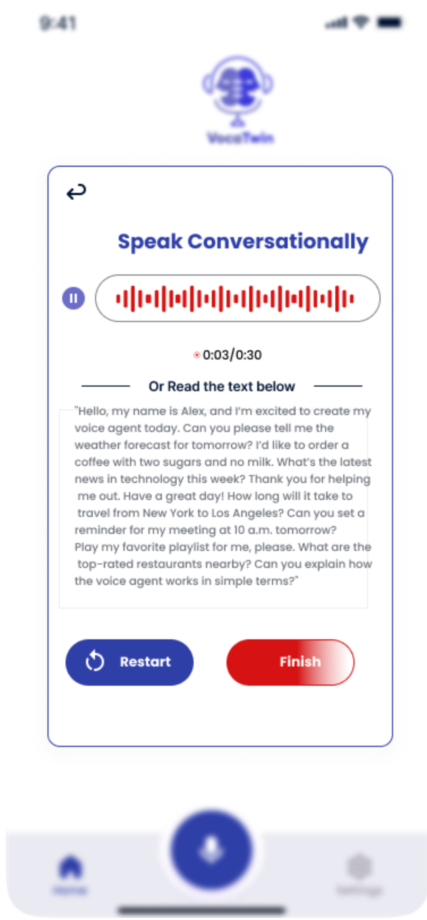
When you tap the big microphone button from the Home Page, you'll be taken to the Voice Recording screen — this is where you create your unique voice avatar.

Steps to Record Your Voice:

1. Tap the Record button to start speaking.
2. Speak naturally or read the sample text provided below the waveform.
3. You'll see a timer counting up to 30 seconds that's your recording limit.
4. You can pause anytime or restart if you're not happy with the take.

Once you're satisfied:

- Tap the Finish button to complete your recording.

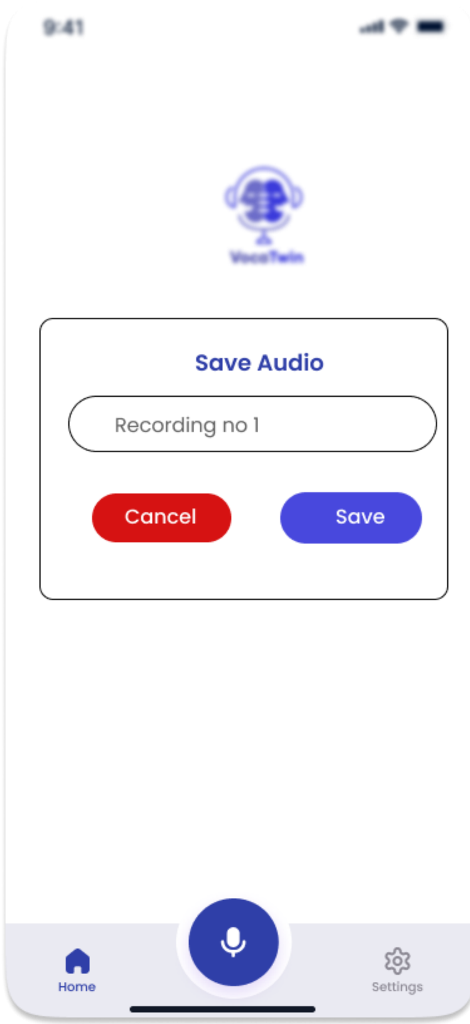


8.7 Save Recording Page (after finishing):

After hitting Finish, you'll be taken to the Save Recording screen.
Here's what you can do next:

1. Preview your recording by playing it back.
2. If needed, tap Re-record to try again.
3. Tap Save & Create Avatar to save your voice.
    - Your voice will now appear in the Recent Voices section on the Home Page.
    - It will also be used to personalize your avatar.

Your voice is now part of Voca-Twin ready to speak for you in your own tone and style.

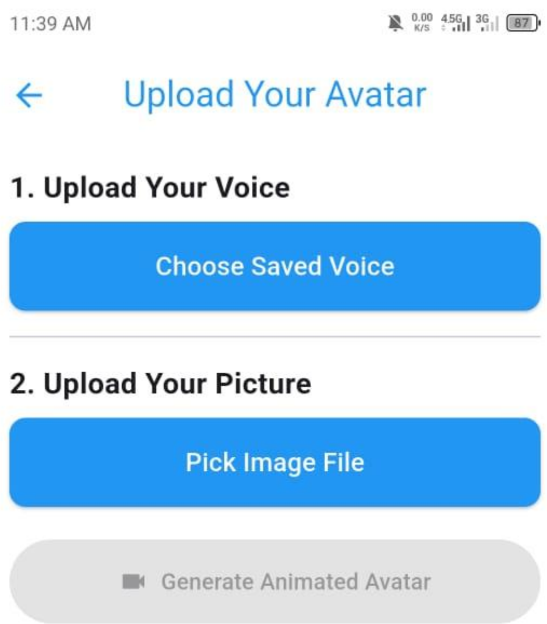


8.8 Choose Avatar

Upon reaching the "Choose Avatar" screen, you will see several built-in avatar options displayed.
Here's what you can do next:

- **Choose your Avatar:** User need to upload his image from phone
- **Proceed to the next step:** Once you have selected your preferred avatar, tap the "Generate Video" button at the bottom of the screen.

After tapping "Generate Video", your chosen avatar will be saved and used within the Voca-Twin application. This is part of the Avatar Creation and Personalization process.

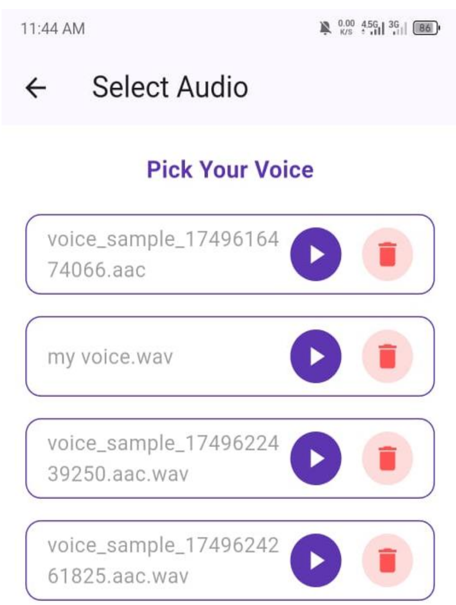


8.9 Pick Your Voice

After recording your voice and selecting your avatar, you will land on the "Pick Your Voice" screen. This is where you choose the specific voice recording you want your Voca-Twin to use for interactions.
Here's what you will see and what you can do:

- You will see a list of your voice recordings. For a newcomer who just finished their first recording, you'll see "Recording 1" listed, likely with a play button next to it.
- **Select your voice:** To confirm that this is the voice you want your avatar to use, tap the "Select Your Voice" button at the bottom of the screen.

Tapping "Select Your Voice" confirms your choice, and this recorded voice will be linked to your personalized avatar



8.10 Text to Speech

This is where you can type messages and have your personalized avatar speak them aloud. Your avatar will use your own cloned voice and its mouth movements will match the speech.
Here's how to make your avatar talk:

- Tap on the message box at the bottom of the screen.
- Type the words you want your avatar to say into the text field.
- Tap the send button to have your avatar speak the message.

Your avatar will then speak the message you typed, using your cloned voice, with synchronized mouth movements.